Information Visualization

Theme 1. Visual Perception and Encoding

Definitions

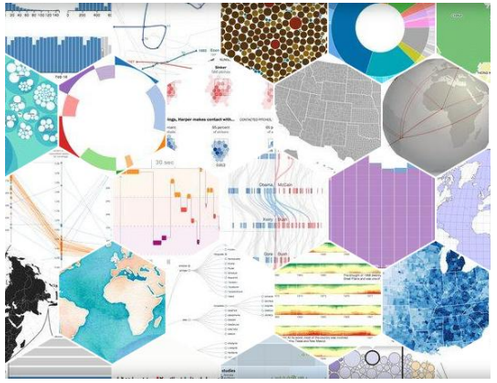
- What is Visualization? Information Visualization? Data Visualization? Visual Analytics?

1.1 Motivation

Visualization (nowadays)







https://d3js.org/

1.1 Motivation

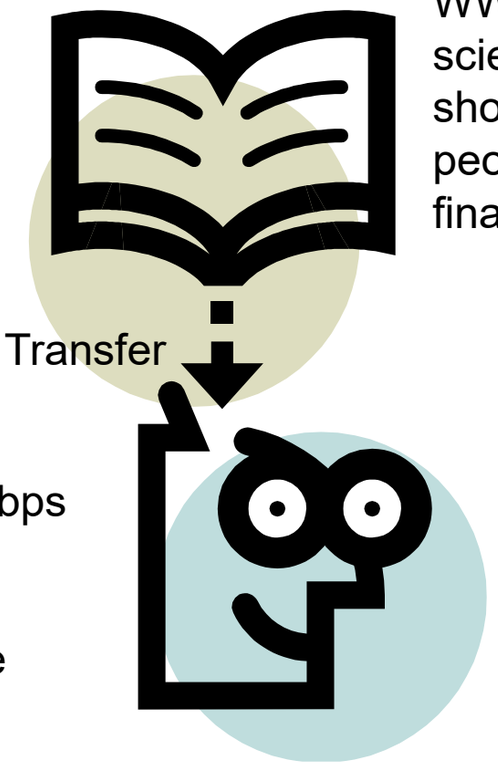
Visualization (nowadays)



https://d3js.org/

1.1 Motivation

- Problem: How to make use of the data?



WWW, books, article, scientific data, biotech, shopping, data on specific people, economics (stock, finance), news, …

Transfer

Vision: 10 Mbps
Ears
Telepathy
Haptic/tactile
Smell
Taste

Inspired by J. Stasko‘s course

http://www.eurekalert.org/pub_releases/2006-07/uops-prc072606.php

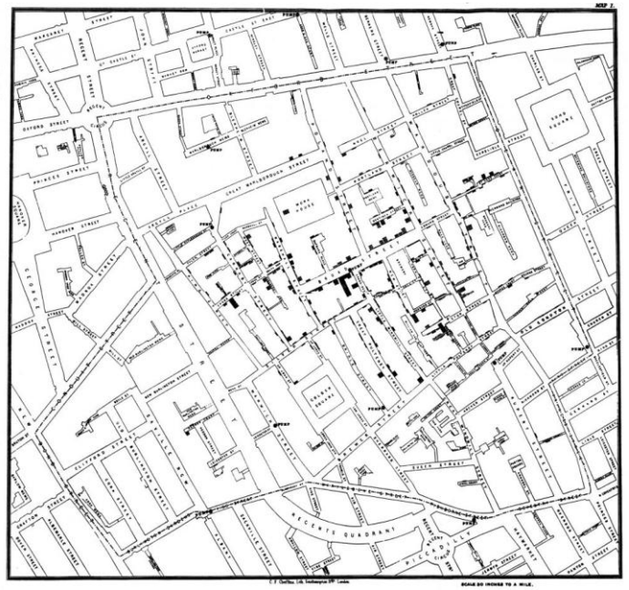
1.1 Motivation

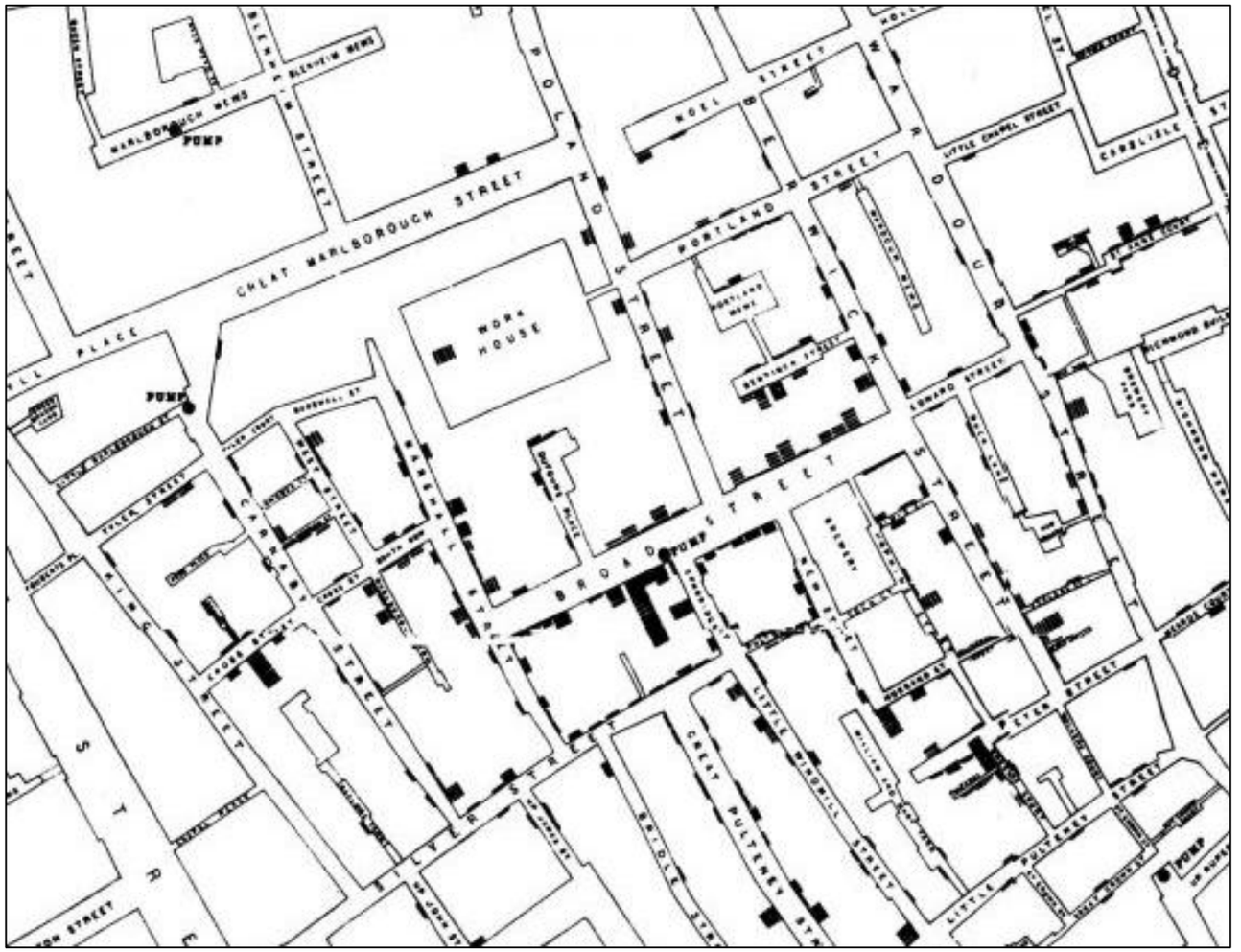
- Human Vision
  - Sense with highest bandwidth
  - Fast, parallel
  - Pattern recognition
  - Preattentive perception (ca. 60ms latency time)
  - Many people think visually

- Challenge
  - Transform the *data* into *information* (understanding, insight) and make them useful (available, analyzable) to people!

1.1.1 Classical Examples

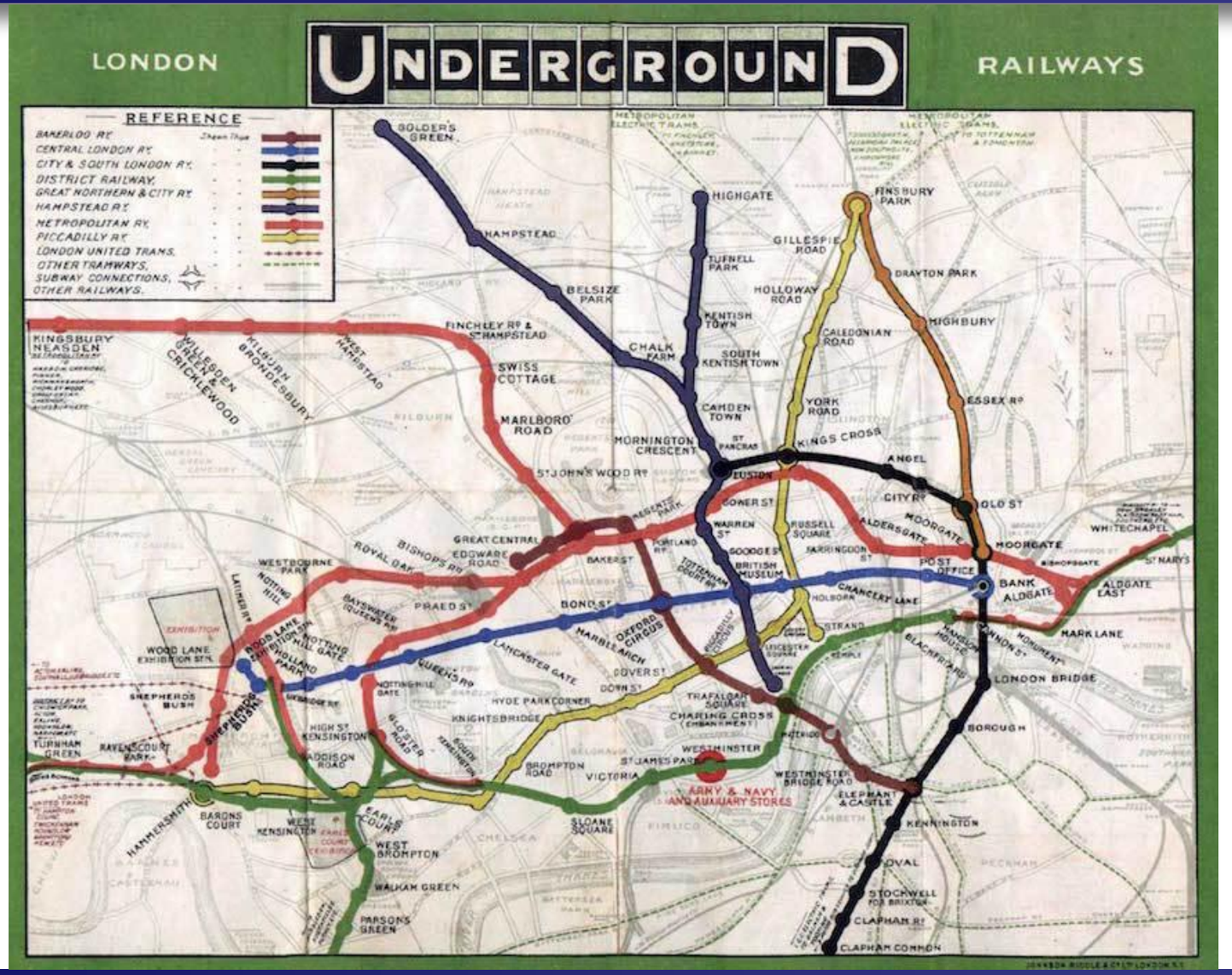
- Cholera epidemic, 1845
- John Snow





[Spe:7f]

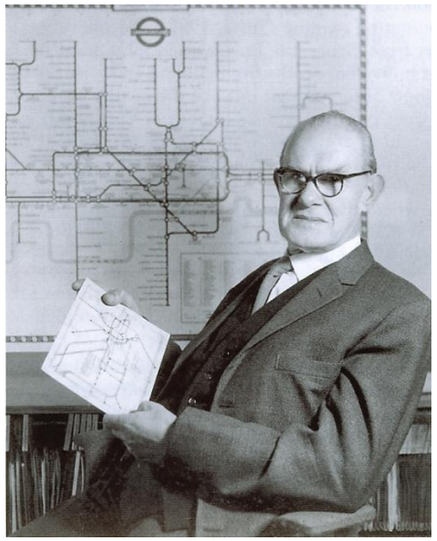
1.1.1 Classical Examples



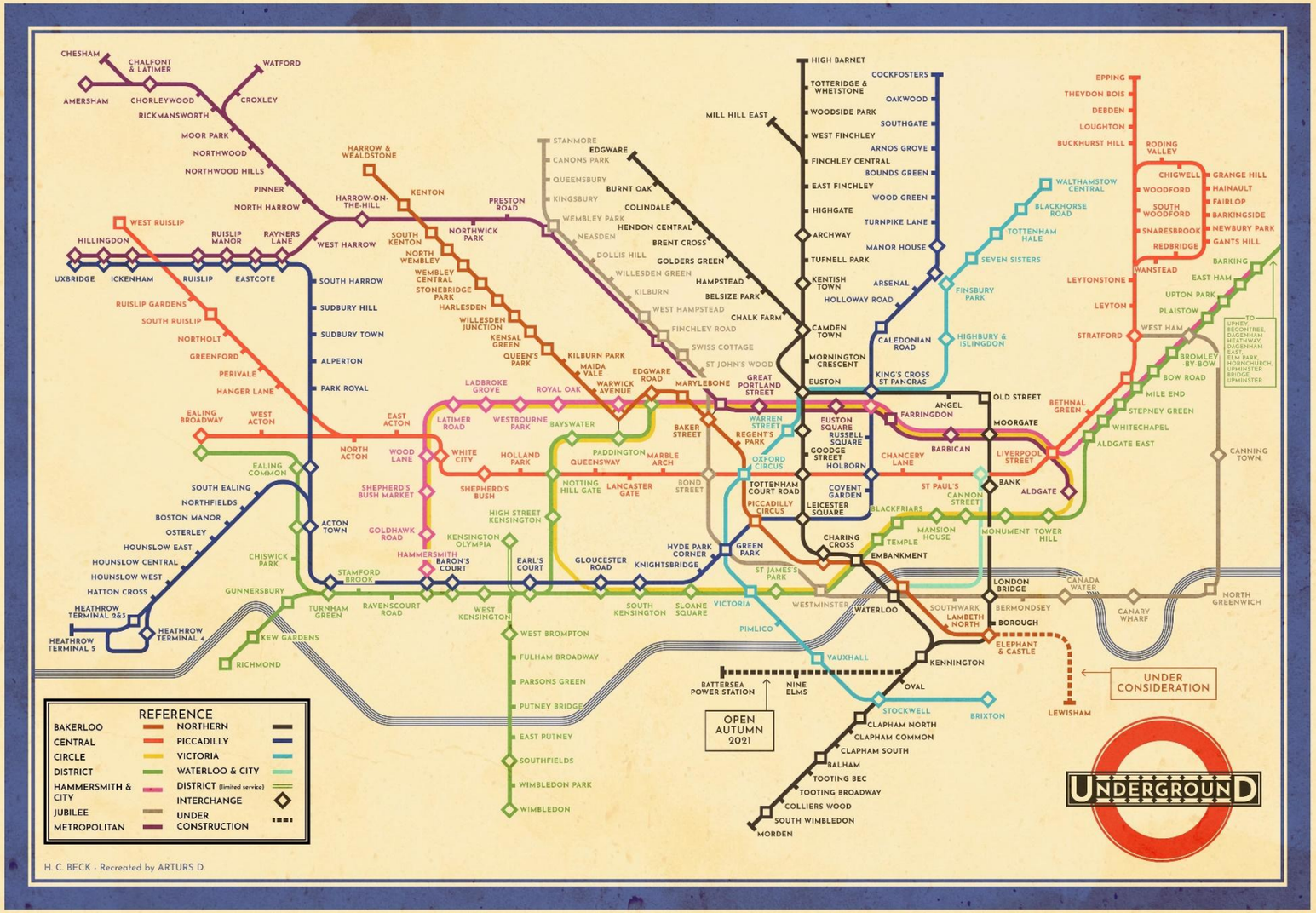
[Spe:7f]

1.1.1 Classical Examples

- Harry Beck‘s Map, 1931



[Spe:2ff]



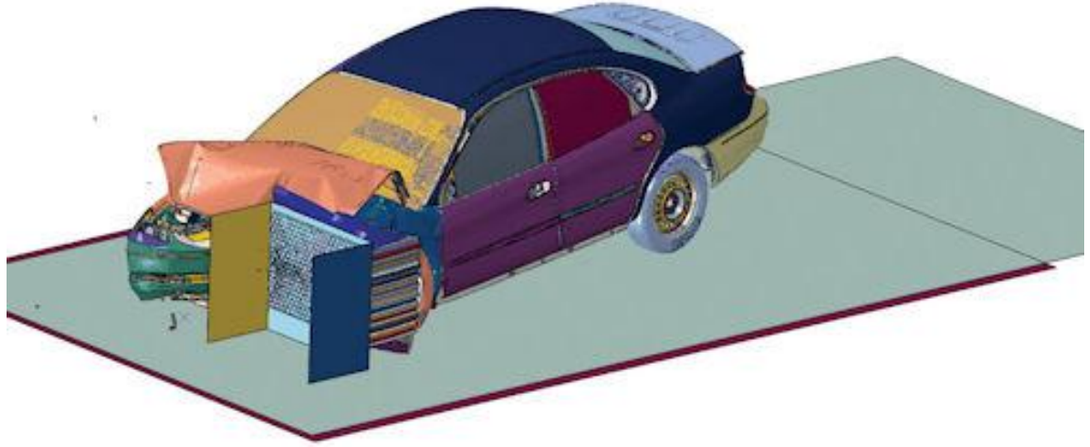
1.2 Definitions

What is a *Visualization*?

- Often defined as a process that produces a graphic, diagram, or an image
- In fact, it is a cognitive process that
  - produces a mental image or a mental model in the human brain
  - supports a better understanding/insight
- "The purpose of visualization is insight, not pictures" [CMS: 6f]
  - Aims of this *insight*: discovery, decision making, and explanation

1.2.2 Information Visualization

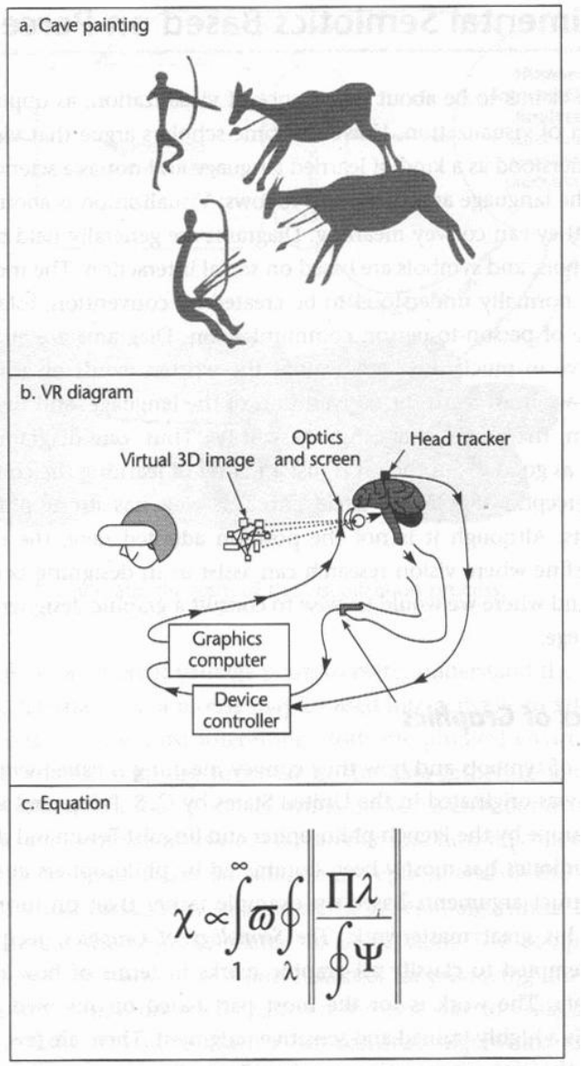
- Information Visualization is **NOT** Scientific Visualization (SciVis)
  - Represents physical phenomena or scientific simulations
  - SciVis uses spatial data **with** physical space
  - 2 or 3-dimensional data
  - Areas
    - Volume Visualization



2.1 Semiotics

- Many visual languages are easy to understand, but not all!
- What are the reasons?
- It seems that there are many cultural and social reasons
  - The following symbol probably makes no sense within an isolated clan in South America



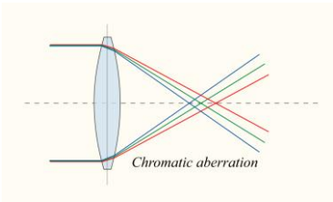


2.2.1 Features

Chromatic Aberration

- Different wavelengths of light are focused at different distances within the eye. In order to refract red (~480 nm) and blue (~640 nm) equally, the human eye needs a correction by a lens of 1,5 diopters. Therefore, a blue text on black background appears diffuse beside white or red text

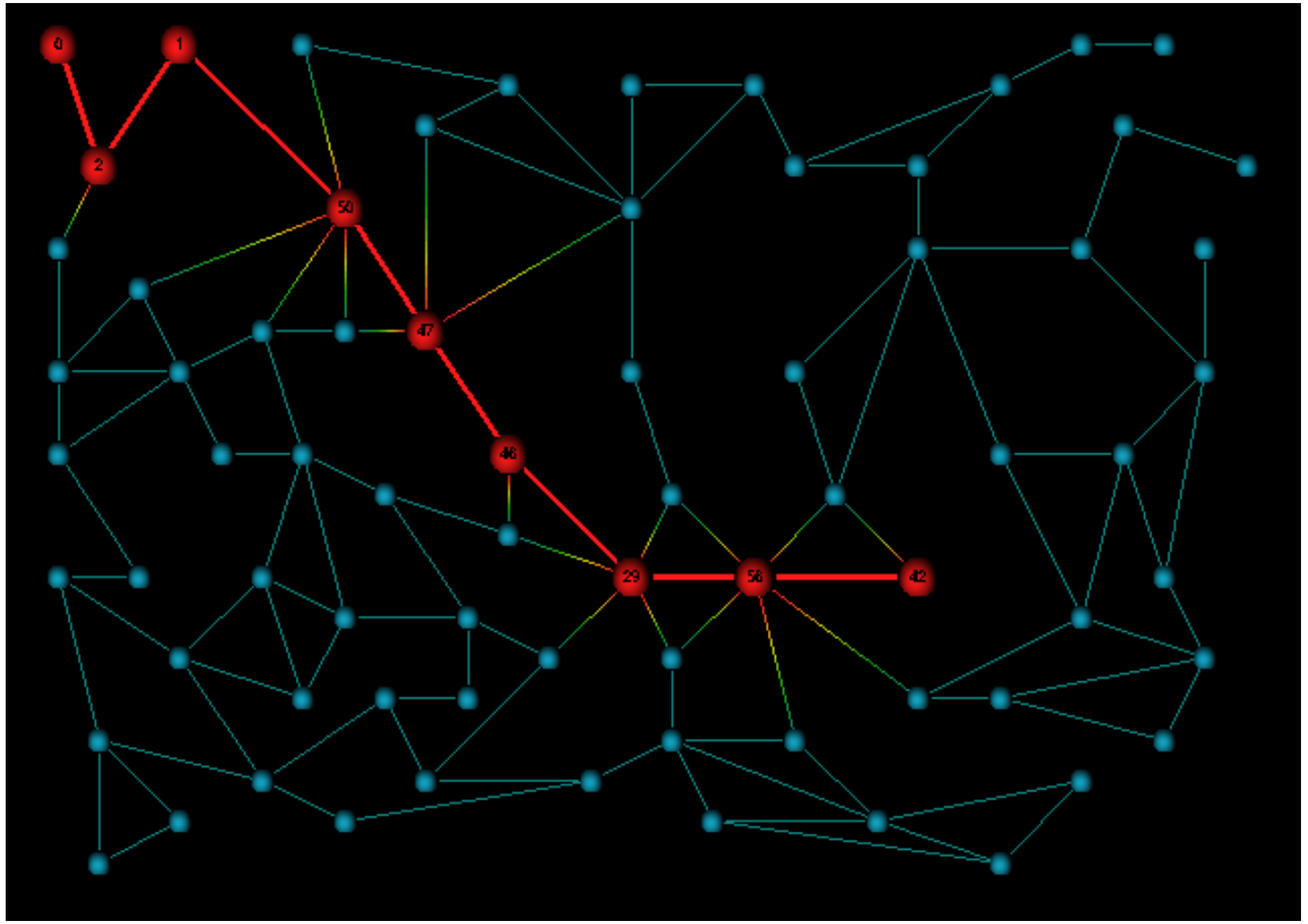
Most people see the red
Closer then the blue,
But some see the
Opposite effect!



- For 60% of observers, the red appears closer, for 30% the blue appears closer and 10% see the colors in the same plane

2.2.1 Features

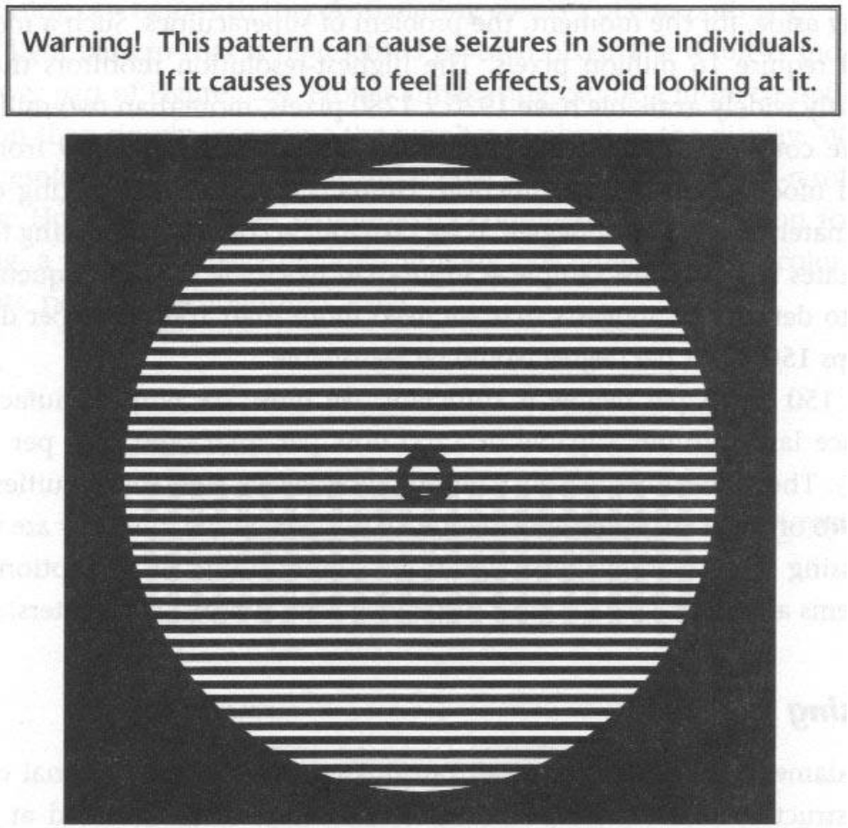
- Application: Focus on specific areas or paths



https://www.cg.tuwien.ac.at/research/publications/2002/Wallisch-2002-INF/TR-186-2-02-08Paper.pdf

2.2.1 Features

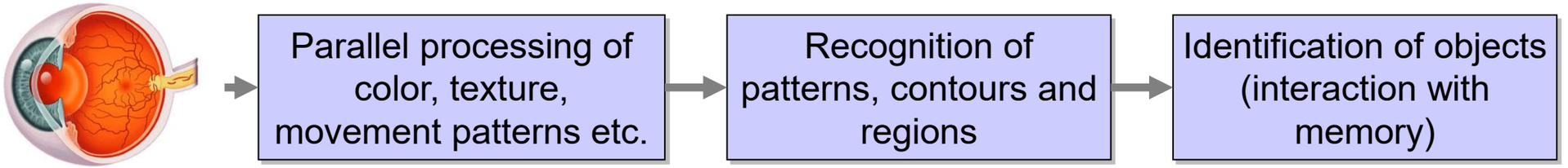
- **Visual Stress**



  - Such patterns can be dangerous and should be avoided in visualizations!!!

2.3 Process Model

- Model according to Colin Ware [War:20ff]
  - Three-level Perceptual Process
    - Level 1: Parallel processing of low-level-properties of a scene
    - Level 2: Pattern recognition (sequential)
    - Level 3: Sequential, target-oriented processing



2.4 Low-level Aspects

- In Level 1 of our process model, several low-level aspects are important, such as
  - Luminance
  - Brightness
  - Contrast
  - Color
  - Texture
  - Gestalt
  - etc.

- Because of time reasons, we only take a very brief look into these concepts. A comprehensive description is given in the book of Colin Ware [War]

[War:69ff]

2.4.2 Grayscale

- Probably not the best way to visually code data, because of contrast problems
  - As an example, we could take a closer look on the population density in a thematic map
  - In general, color scales are better suited for such cases

- Crispening-Effect



The 4 stripes' colors are absolutely identical!

2.4.3 Color

- The meaning of colors was and is important in daily life. Probably, colors were more important in the past thousands of years

Finding cherries





2.4.3 Color

- Note that it is difficult to distinguish colors in larger areas. For small colored areas, we have no possibility of differentiation



Color Categorization

- Only 8 colors + white can be safely used for categorization

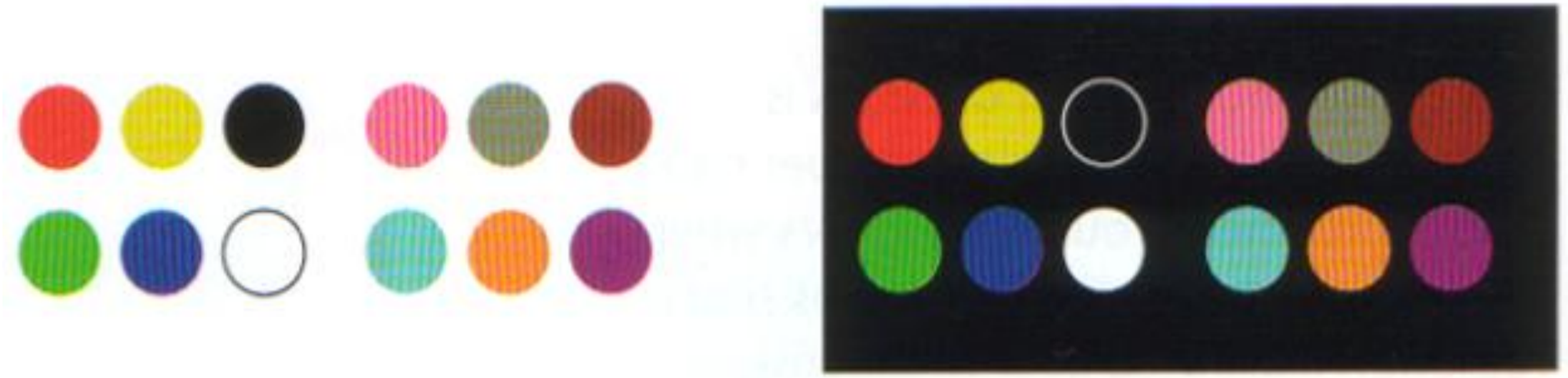


The results of an experiment in which subjects were asked to name 210 colors produced on a computer monitor. Outlined regions show the colors that were given the same name with better than 75% probability.

2.4.3 Color

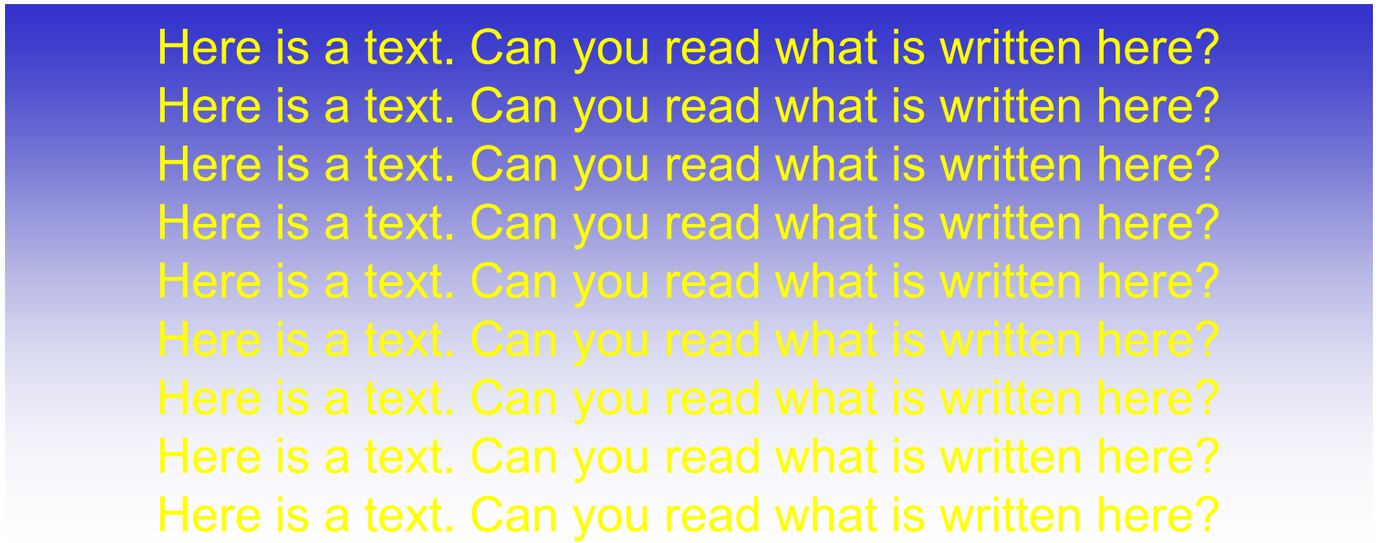
<u>Application:</u> Colors for coding nominal data

- Ware suggests 12 different colors which are displayed here on two different backgrounds



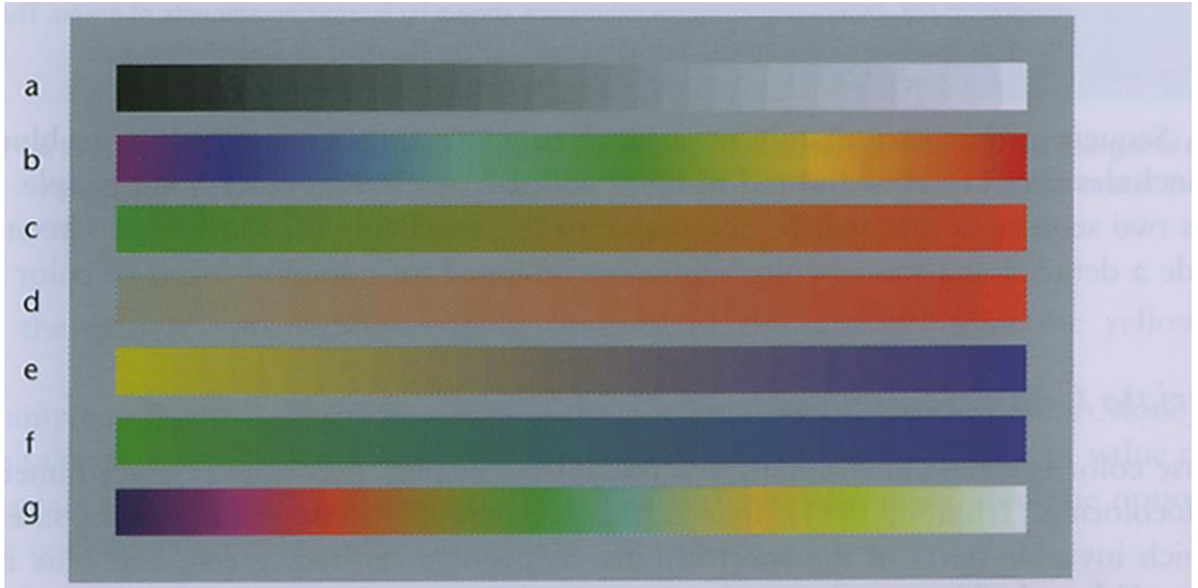
Contrast of Brightness

- Very important for foreground and background colors



Application: Colors for sequence coding

- It is not so easy to find the right sequence, especially if you have to consider color blindness

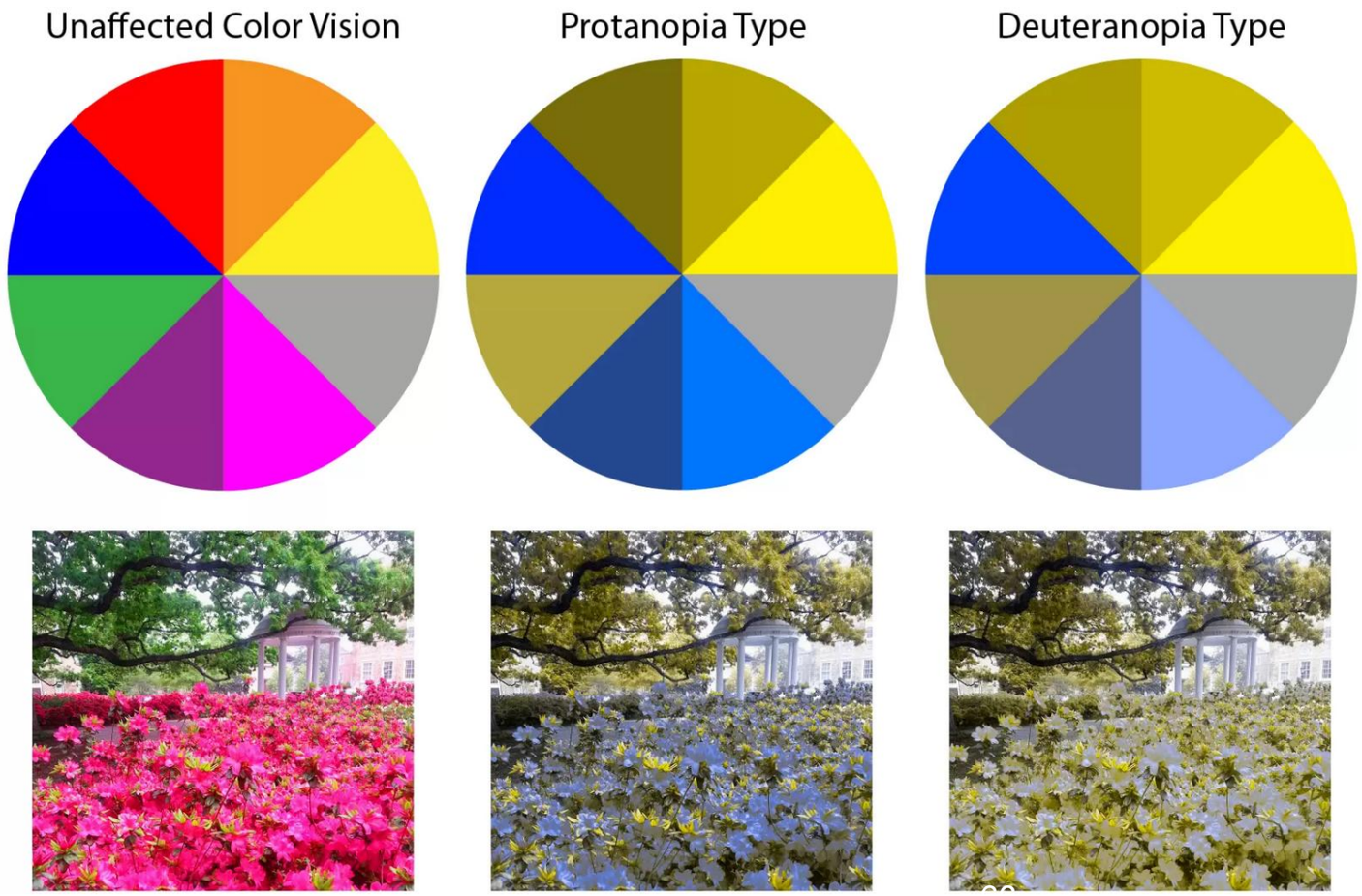


Seven different color sequences: (a) Gray scale. (b) Spectrum approximation. (c) Red–green. (d) Saturation. (e) and (f) Two sequences that will be perceived by people suffering from the most common forms of color blindness. (g) A sequence of colors in which each color is lighter than the previous one.

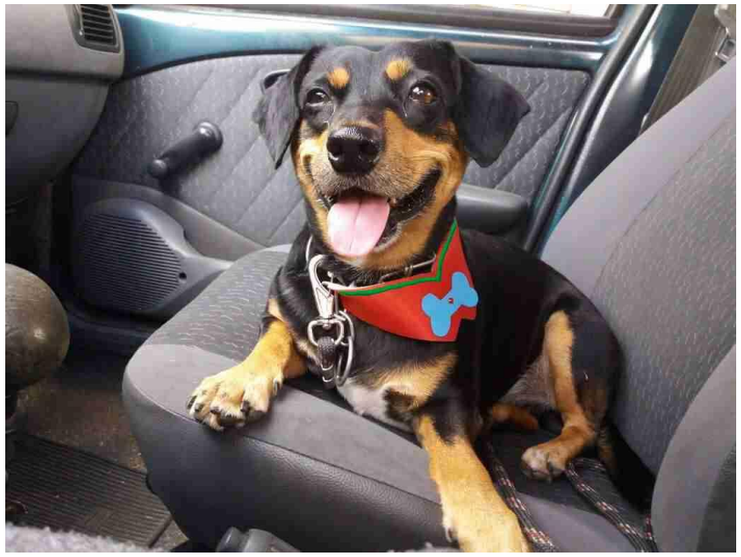
2.4.3 Color

- Color blindness is the difficulty in distinguishing different colors.
- The most common types of color blindness are protanopia and deuteranopia, which are characterized by difficulty in distinguishing **green** from **red**.
- It is usually a common problem that generally affects more men than women.

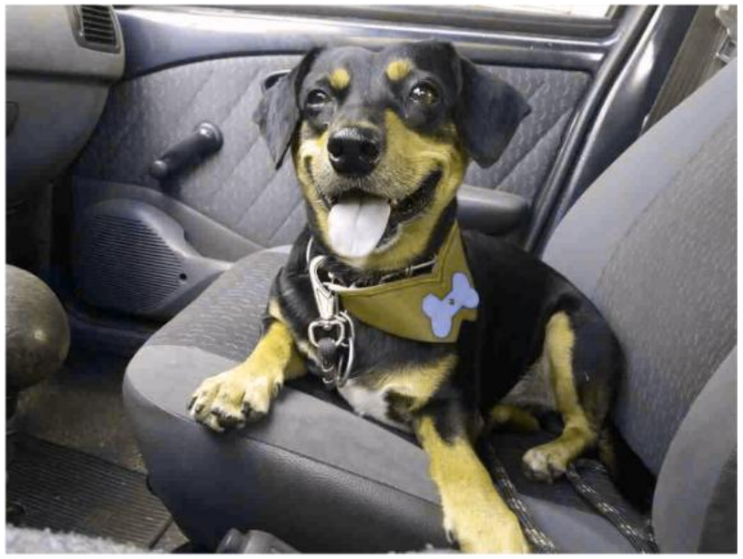
2.4.3 Color


Unaffected Color Vision
Protanopia Type
Deuteranopia Type

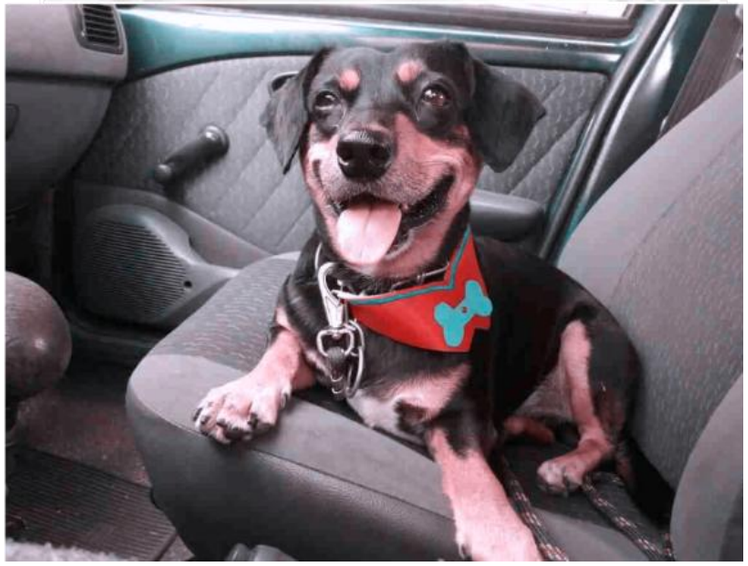
2.4.3 Color



Unaffected Color Vision



Protanopia



Tritanopia

Website to simulate different types of color blindness:

https://www.color-blindness.com/coblis-color-blindness-simulator/

2.4.3 Color

- In some cases, rainbow color map is not ideal
- It is based on the colors in the light spectrum



  - Widely used and people like it, but can be *misleading* (or not?)

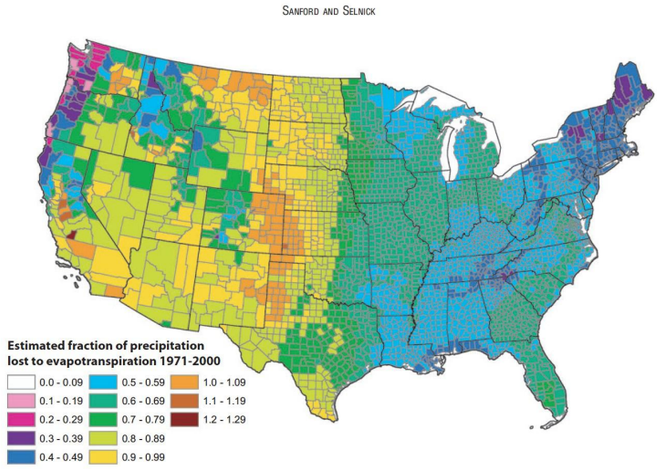


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (*P*) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/*P* were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

https://eagereyes.org/basics/rainbow-color-map

2.4.3 Color

https://eagereyes.org/basics/rainbow-color-map

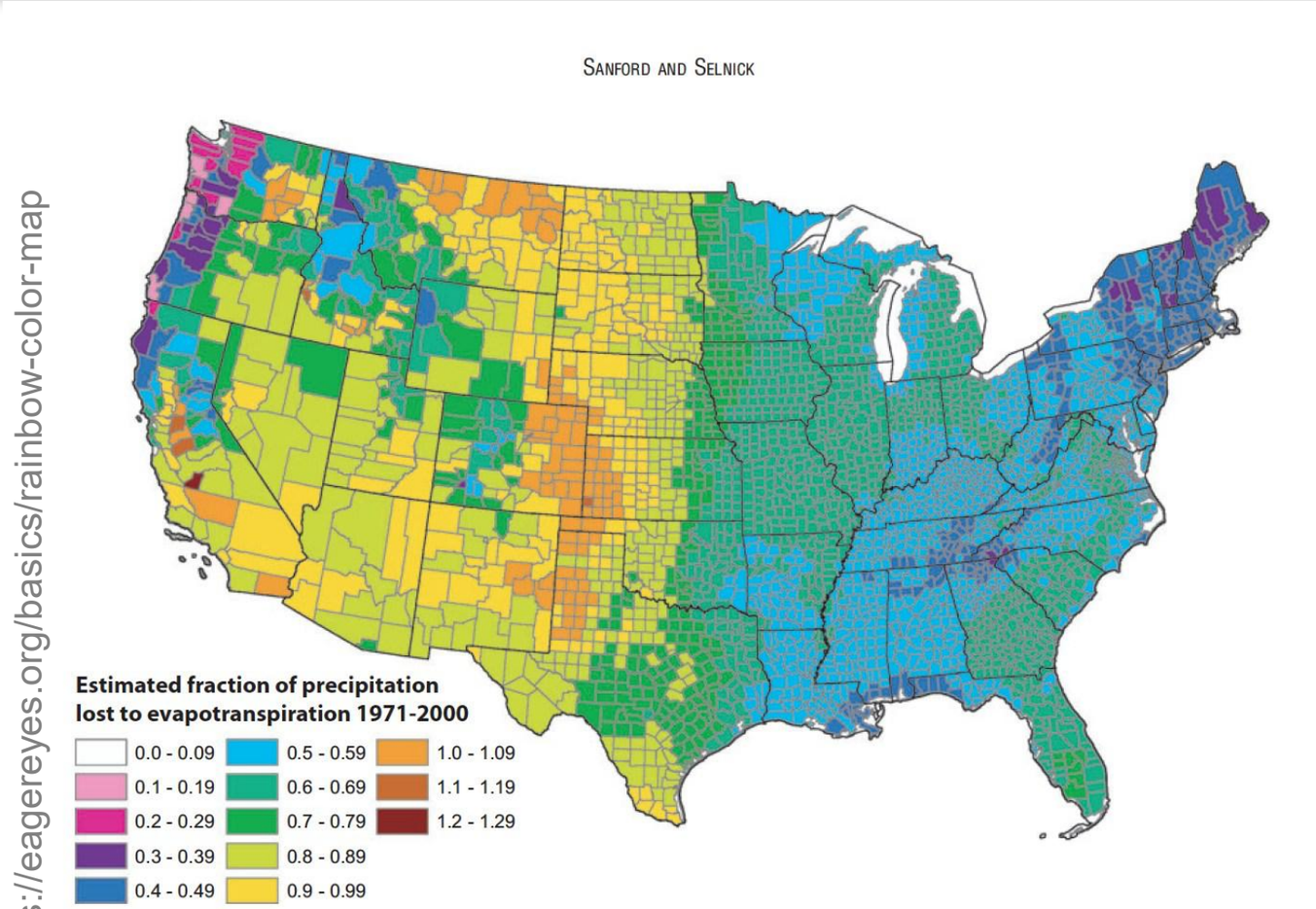


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Actually, there is NO big change as indicated by the vertical borderline in the middle!

2.4.3 Color

- What about converting the rainbow scale into the gray scale?

https://eagereyes.org/basics/rainbow-color-map

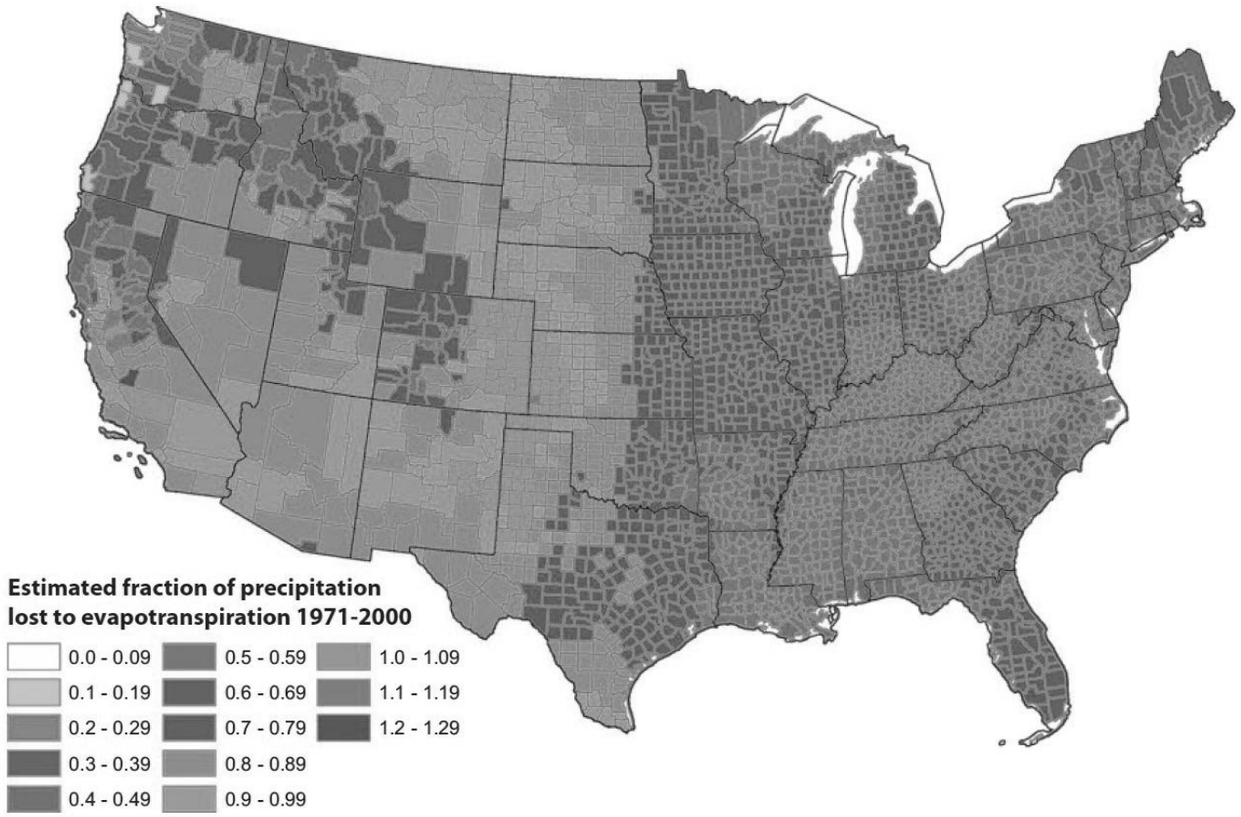


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2.4.3 Color

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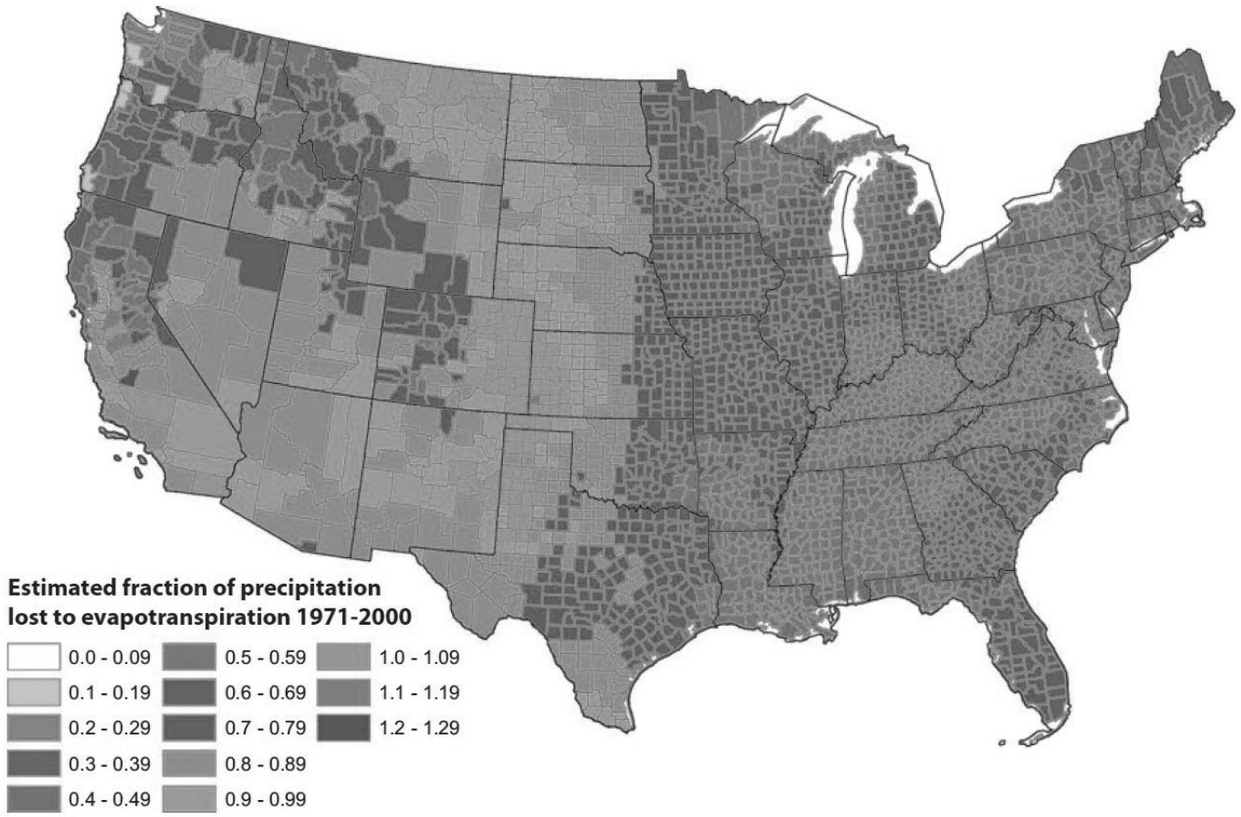
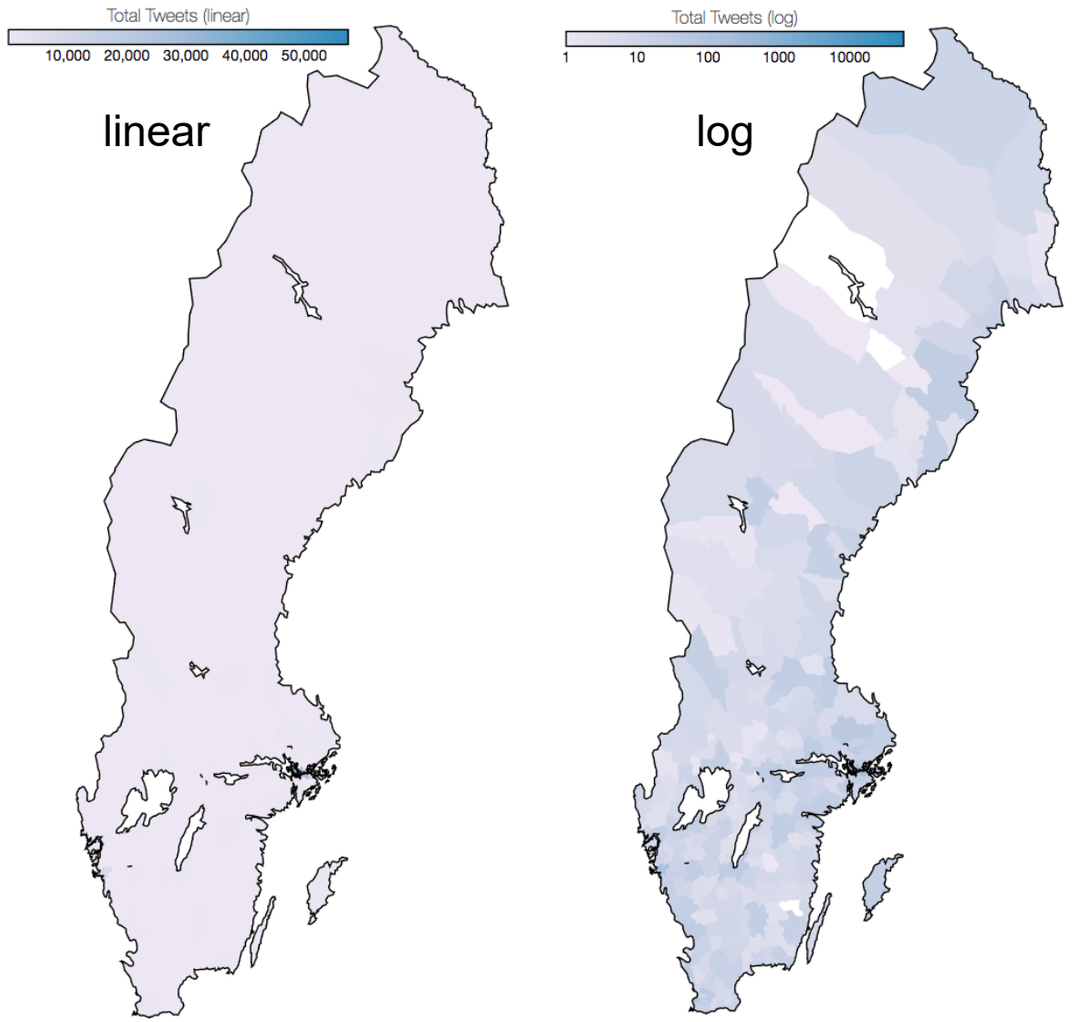


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There is no linear order in relation to scale (e.g., from brighter to darker), even though this could be meaningful on sequential data

2.4.3 Color

- Influence of scale in the color mapping



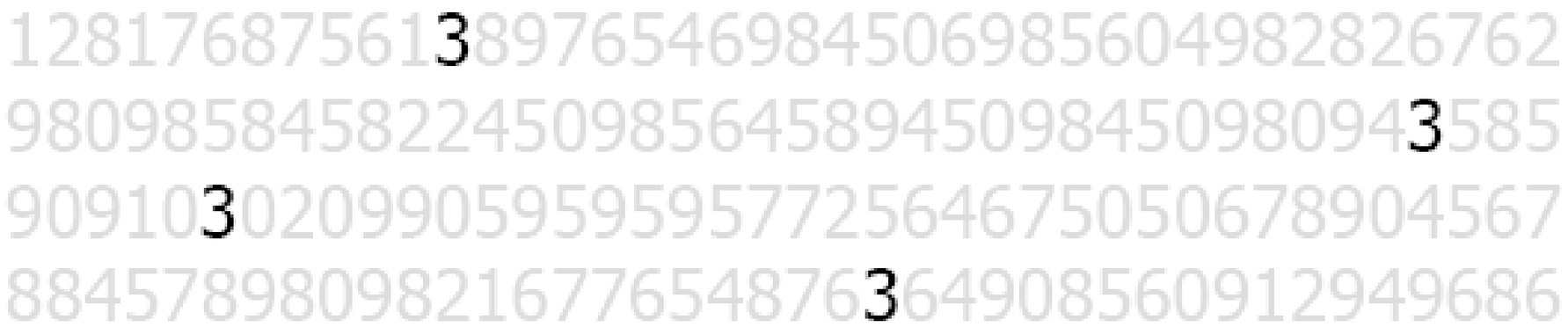
2.5 Preattentive Processing

- Count the number **3**

```
12817687561389765469845069856049828267 62
98098584582245098564589450984509809435 85
90910302099059595957725646750506789045 67
88457898098216776548763649085609129496 86
```

2.5 Preattentive Processing

- Count the number **3**



2.5 Preattentive Processing

- Special colors or shapes stand out their graphical environment (without any search efforts!!!)
- *Preattentive Processes* are the basis of this phenomenon, i.e., they determine which visual objects are in our focus

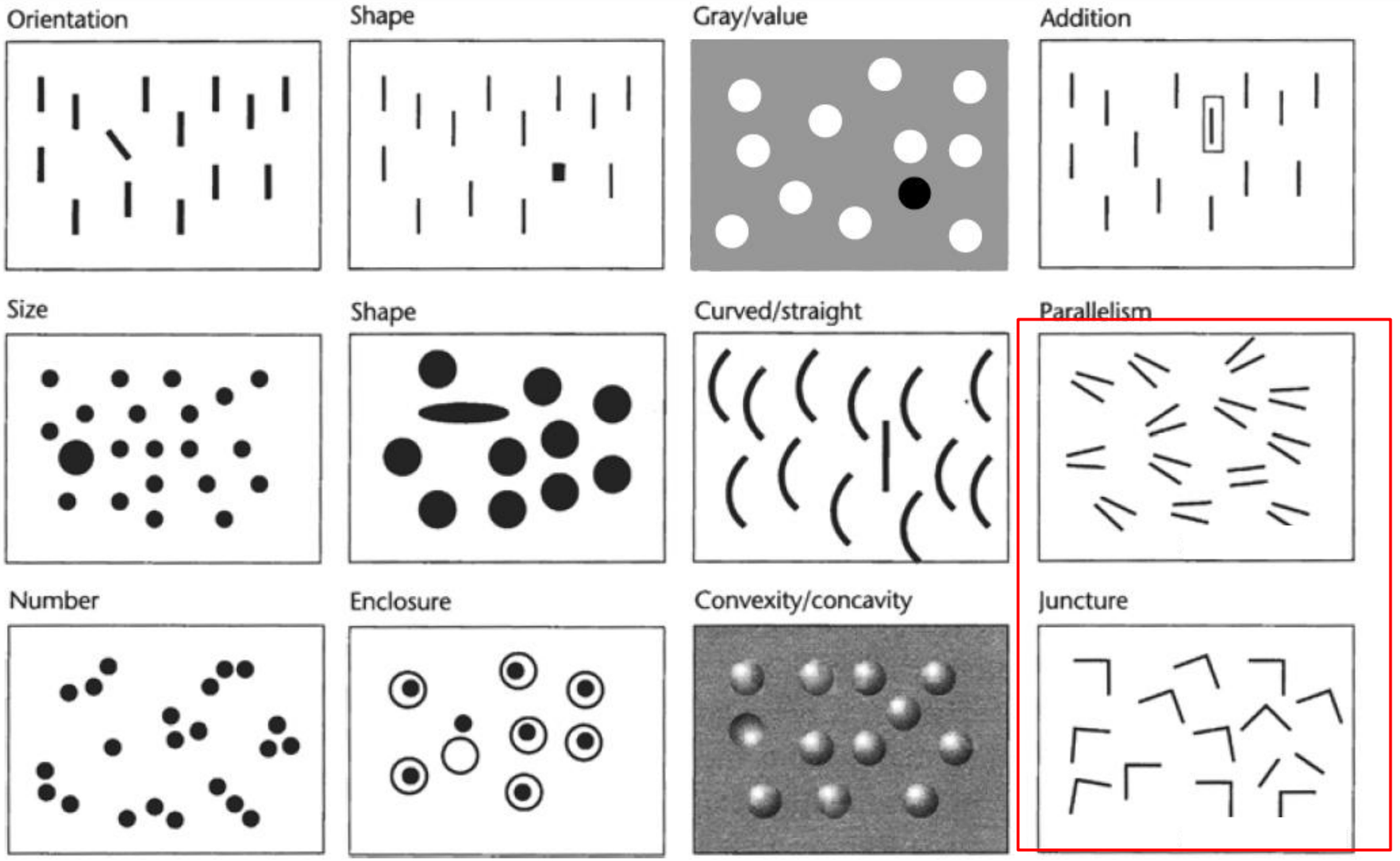
  → Very important for InfoVis
- Preattentive processes happen in Level 1 of our process model
  - Low-level
  - Parallel
  - Fast (≤ 40-60 ms)

[War:149ff]

2.5.1 Preattentive Features

- Form
  - Line orientation
  - Line length
  - Line width
  - Size
  - Curvature
  - Spatial grouping
  - Added marks
  - Blur
- Color
  - Hue
  - Intensity
- Motion
  - Flicker
  - Direction of movement
- Spatial Position
  - 2D position
  - Convex/concave shape from shading

2.5.1 Preattentive Features



Does not work!

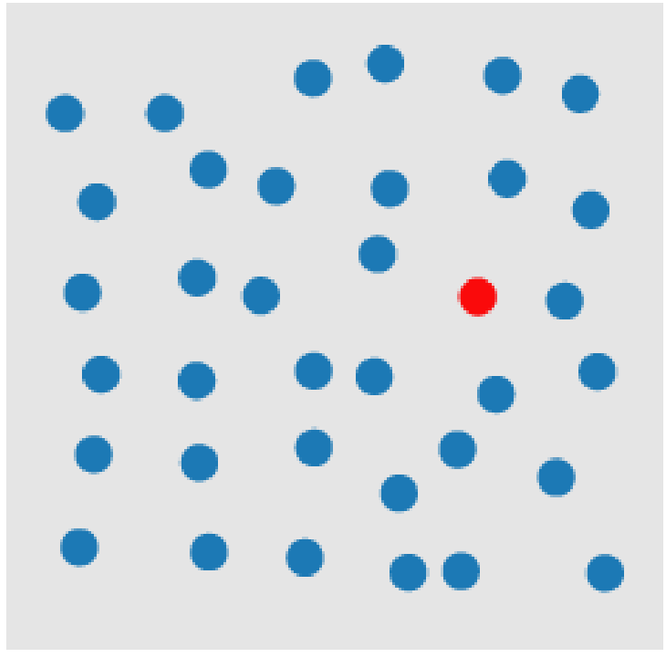
2.5.1 Preattentive Features

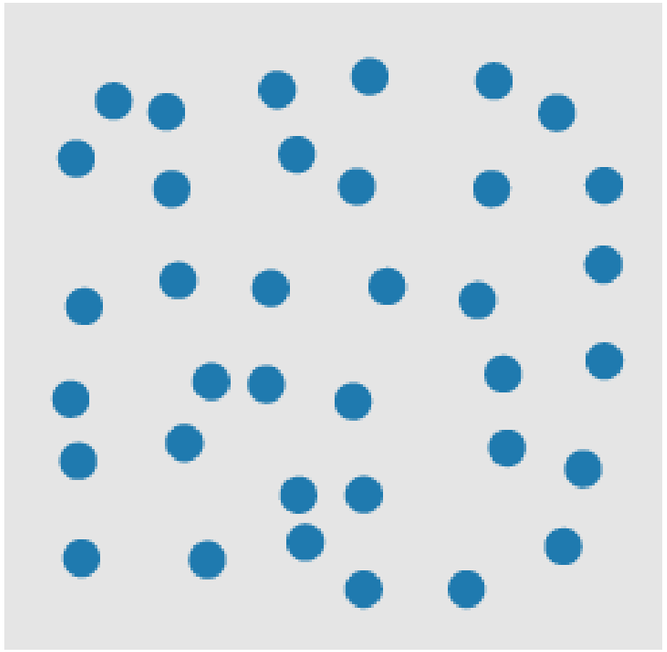
- Studies showed two important factors if we have a preattentive feature
  - The degree of distinction of the target object from the distractors AND
  - the degree of distinction of the distractors among each other

- Furthermore, *combinations of preattentive features* are very dangerous because they often lose their property of preattentiveness …

2.5.1 Preattentive Features

- Do you see a red dot?
- Preattentive Feature: *Color*



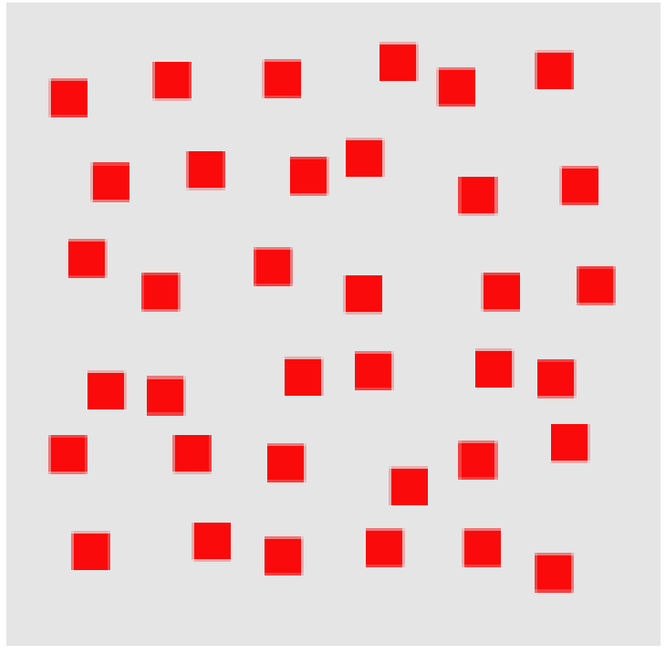


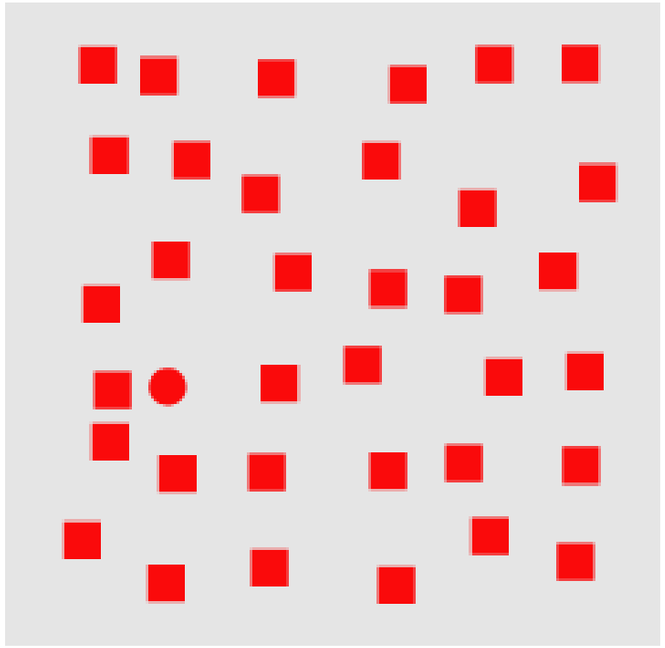
Preattentive!

http://www.csc.ncsu.edu/faculty/healey/PP/

2.5.1 Preattentive Features

- Do you see a red dot?
- Preattentive Feature: *Shape*



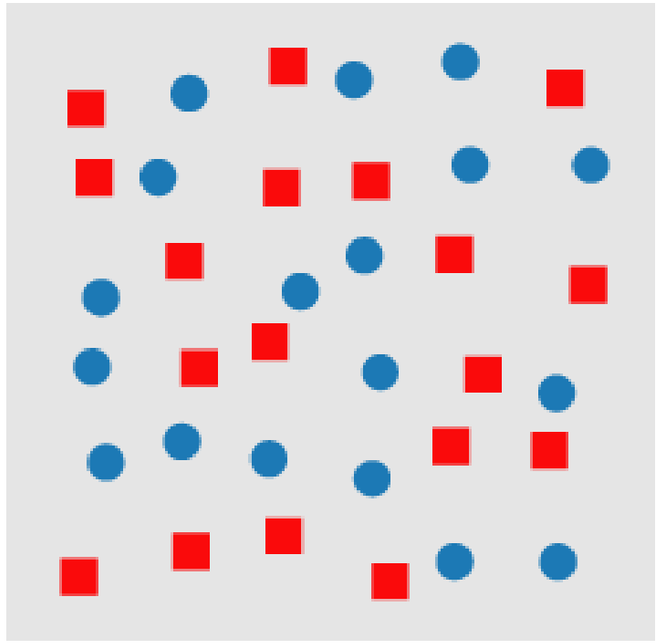


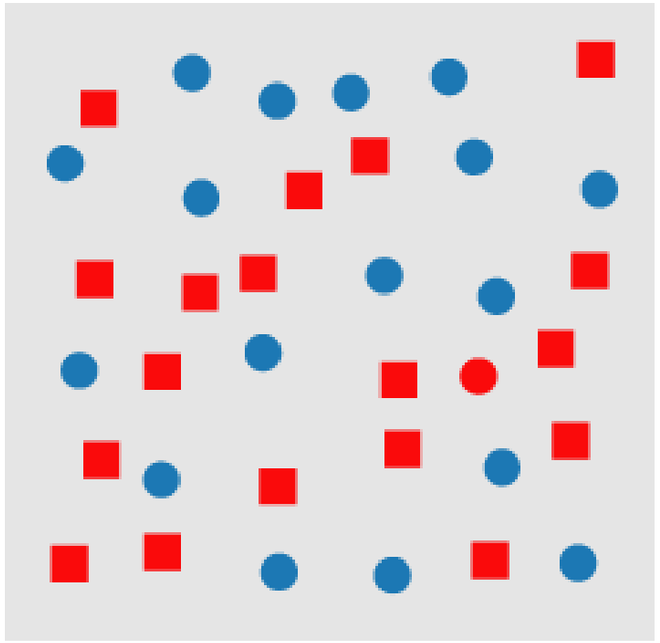
Preattentive!

But it is harder than color

2.5.1 Preattentive Features

- Do you see a red dot?
- Preattentive Features: *Color* and *Shape*

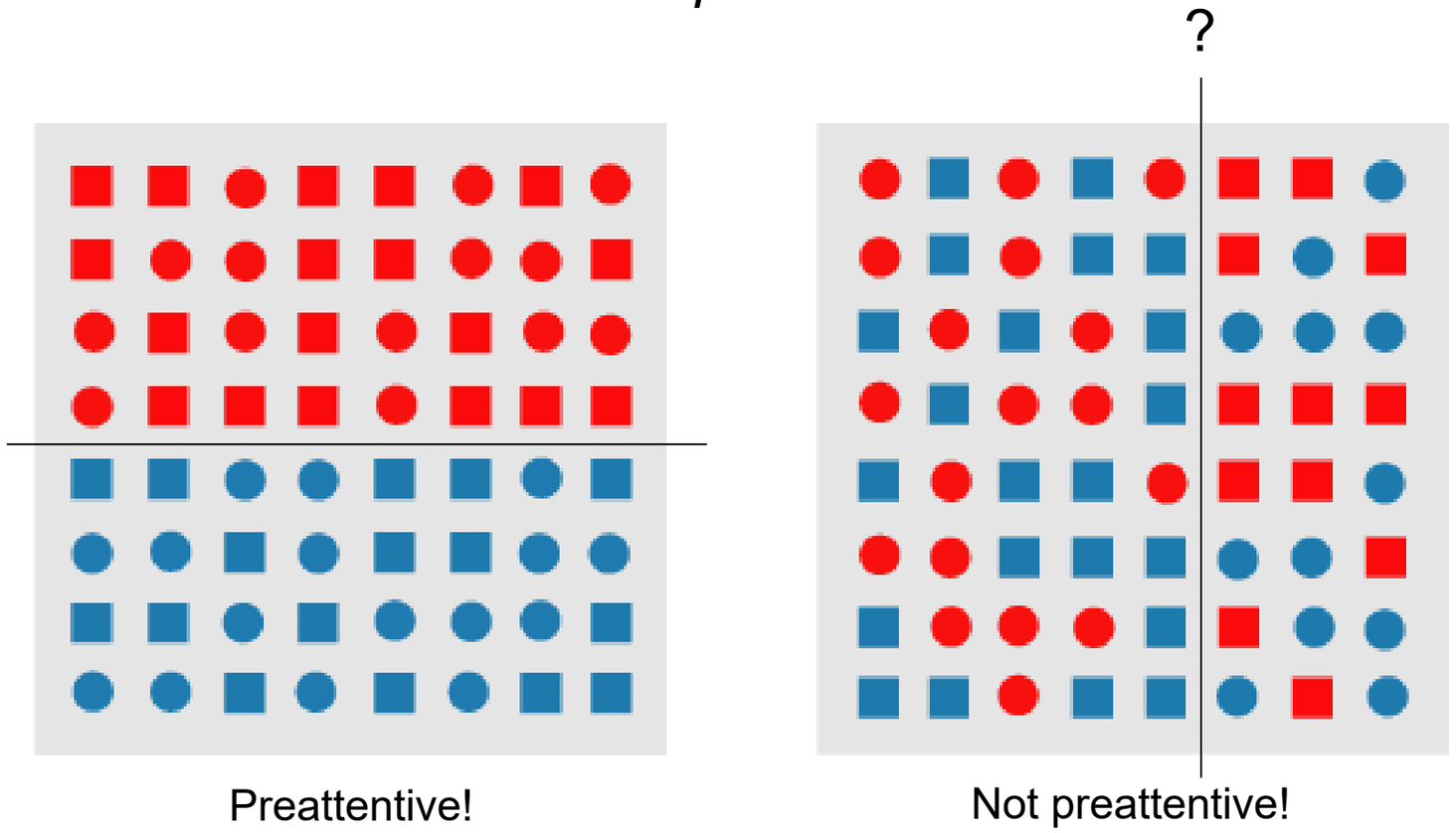




Not preattentive! (Sequential search needed)

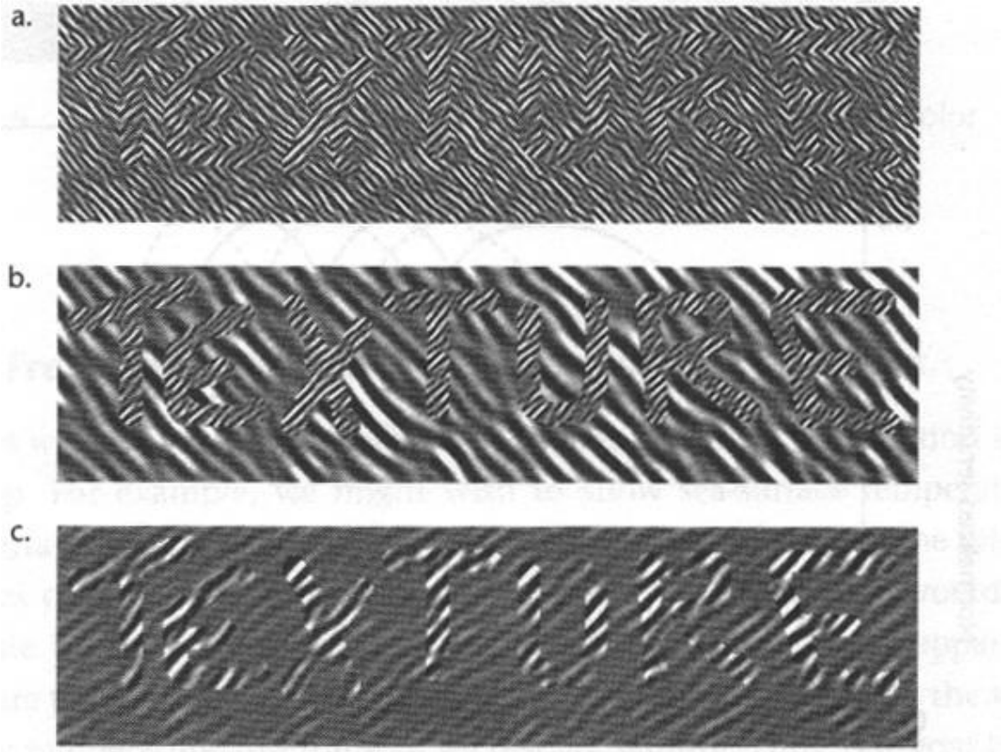
2.5.1 Preattentive Features

- Where is the border?
- Preattentive Features: *Color* and *Shape*



2.6 Textures and Glyphs

- We discuss *textures* only briefly. They are discussed in detail in the book of Ware [War:161ff]
- Level 1 in our process model, i.e., low-level perception
- The world of textures is certainly as rich as the world of colors



The word *TEXTURE* is visible only because of texture differences between the letters and the background; overall luminance is held constant. (a) Only texture orientation is altered. (b) Texture orientation and size are altered. (c) Texture contrast is altered.

2.6 Textures and Glyphs

- **Glyphs** utilize various visual channels and other types of signals, such as symbols and icons.

- They are excellent for summarizing information, such as warnings and illustrations of behaviors and feelings.

- Glyphs are also cultural.



Glyph-based Visualization: Foundations, Design Guidelines, Techniques and Applications, EUROGRAPHICS 2013

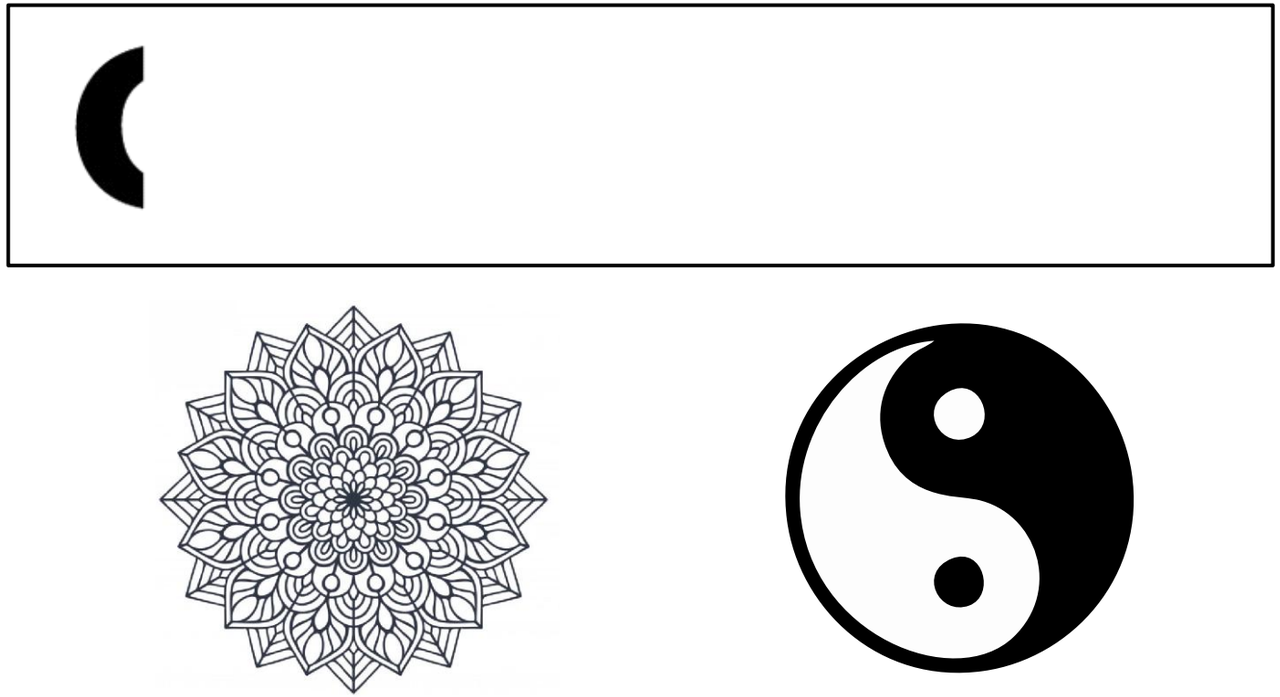
2.7 Gestalt Laws

- Gestalt school of Psychology (founded in 1912 in Germany)
- Gestalt laws are robust rules that describe the way we perceive patterns (Level 2 in our process model)
- A set of 9 laws that study visual perception
  - We will discuss here in the lecture the most interesting ones

[War:189ff]

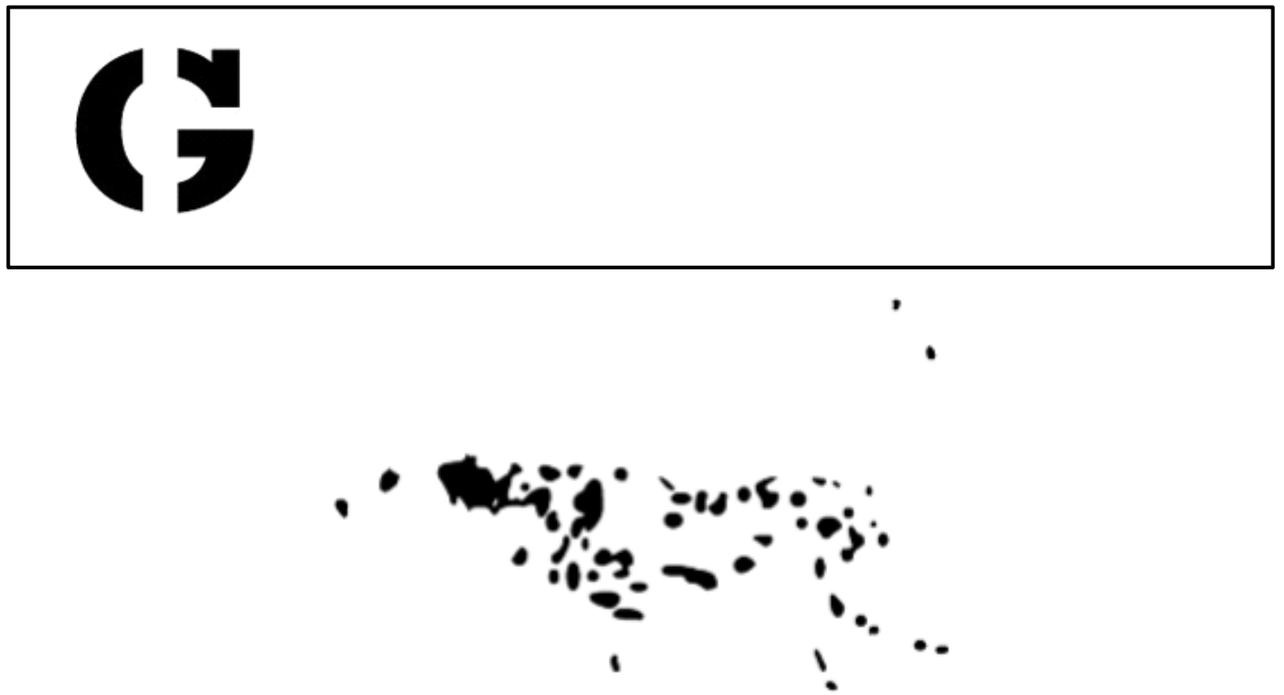
2.7 Gestalt Laws

- Symmetry: Indicates balance and harmony to create symmetrical and pleasing objects;



2.7 Gestalt Laws

- Closure: The brain “completes” images using our prior knowledge;



2.7 Gestalt Laws

- Closure: The brain “completes” images using our prior knowledge;





2.7 Gestalt Laws

- Proximity: Depending on the proximity of the elements, our brain sees them all as the same element;





2.7 Gestalt Laws

- Continuity: The brain's ability to continue objects even when they are superimposed on others;



2.7 Gestalt Laws

- Similarity: The ability to recognize similar objects that are visually grouped together;





2.7 Gestalt Laws

- Fore-/Background: The ability to separate objects, even if they are visually united;



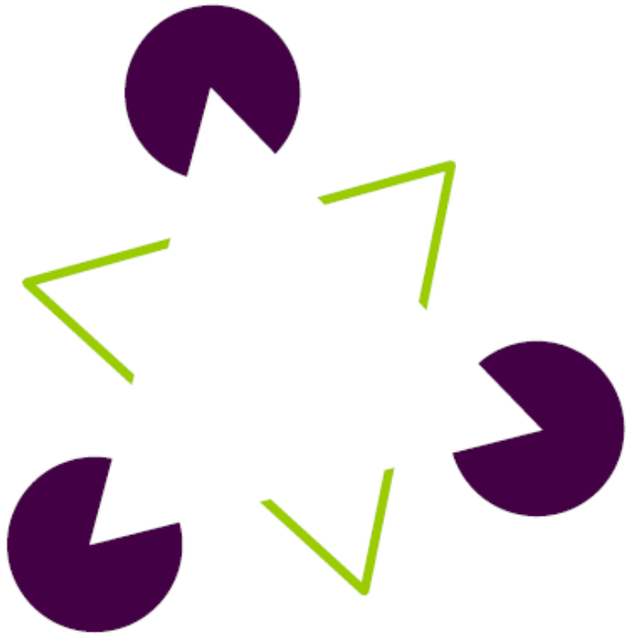
2.8 Static Patterns

- So far, we discussed Gestalt laws in context of static patterns only
- Here, we only discuss 2D patters. 3D patters are presented in the book of Colin Ware in detail [War:227ff]
- Level 2 in our process model
- Among others, *contours*, *diagrams* and maps are static patterns
- A contour is a continuous perceived boundary between regions of a visual image
  - They can be defined by lines, colors, textures, stereoscopic depth, etc.

[War:198ff]

2.8.1 Contours

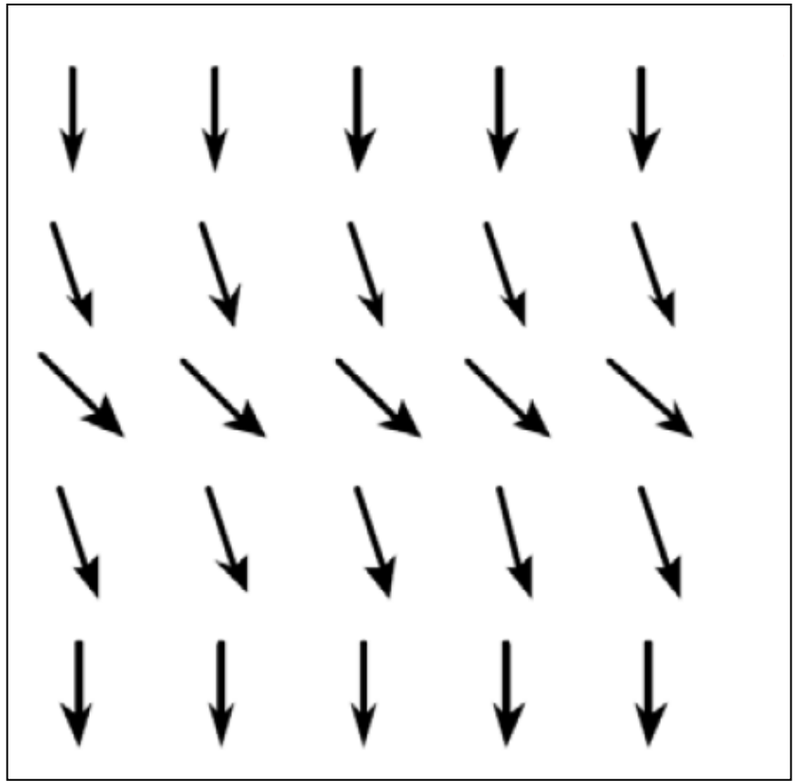
- Illusory Contour

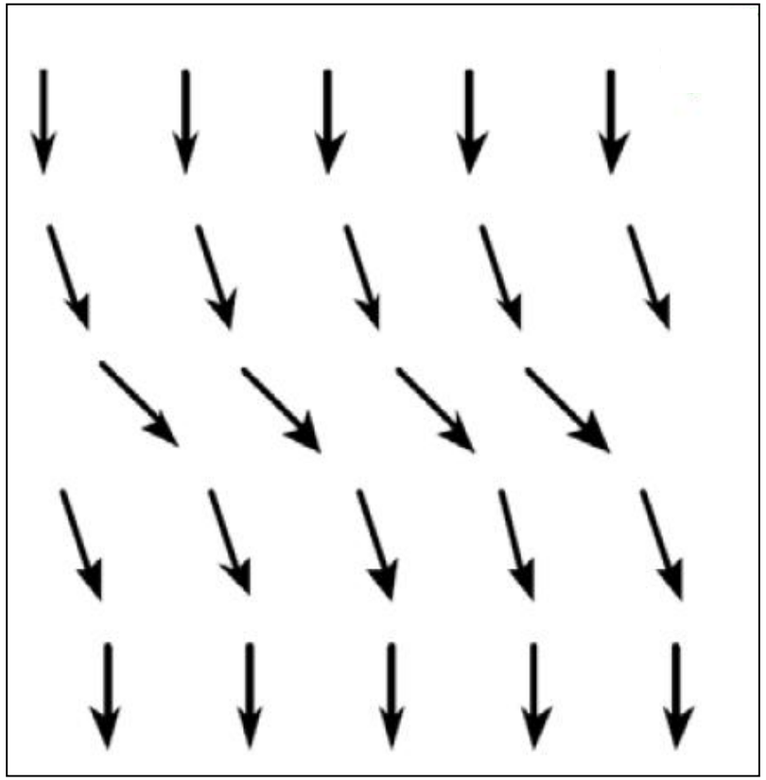


[from H. Hauser's course, 2005]

2.8.1 Contours

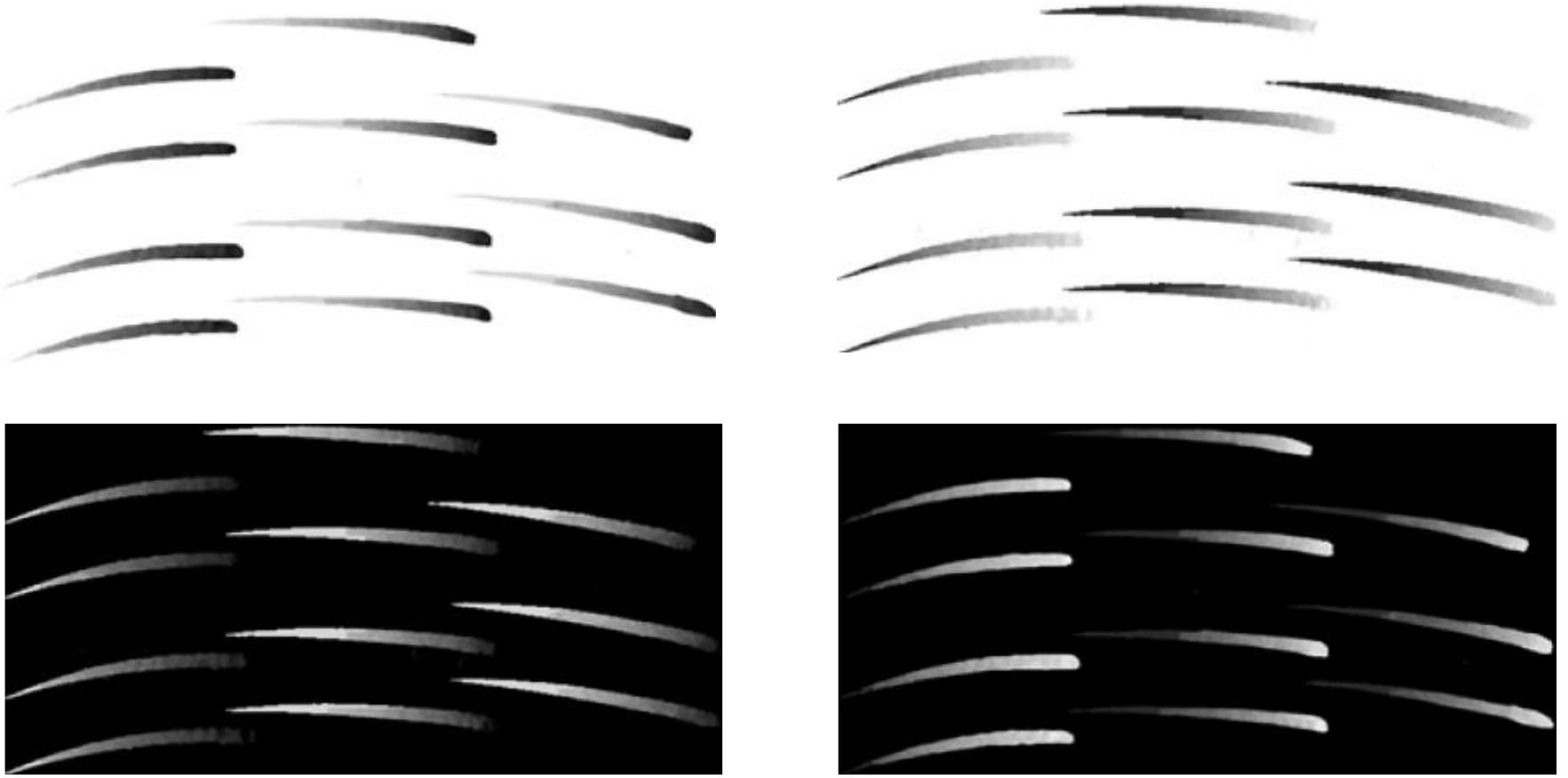
- Perception of a direction
  - Example: Visualization of vector fields





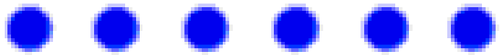
2.8.1 Contours

- Flow direction comparisons (What is the impression of direction?)

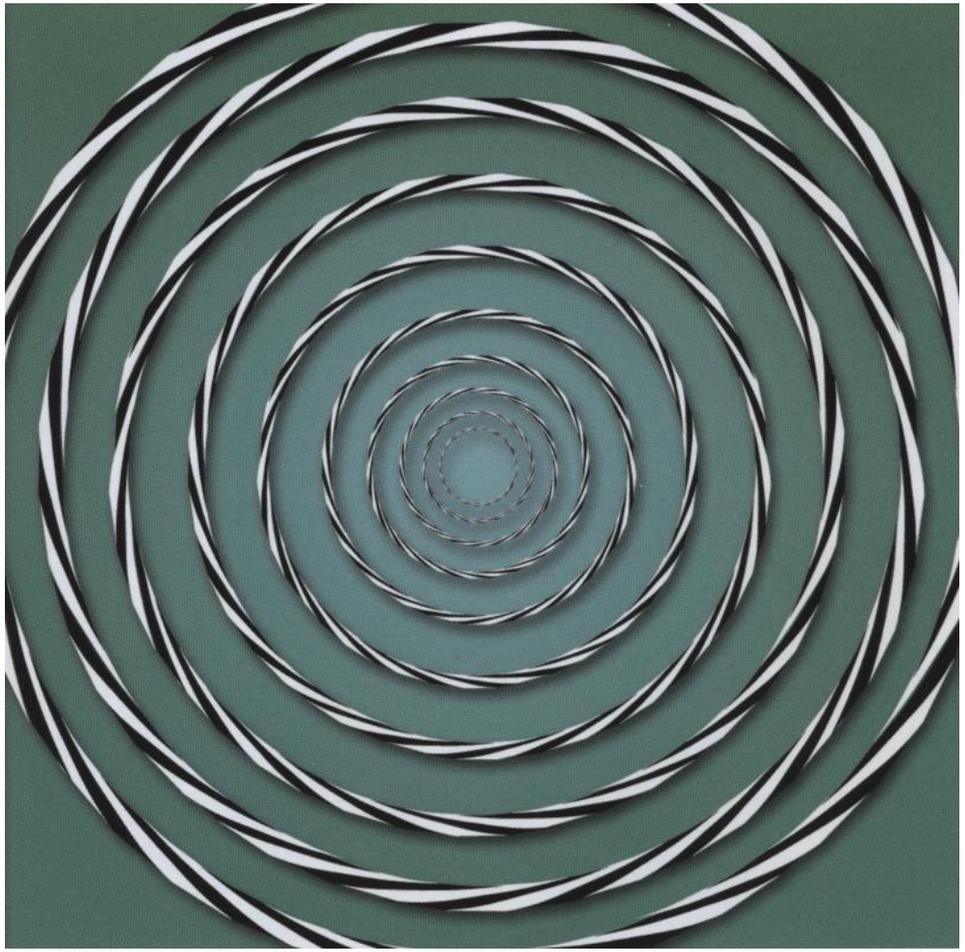


2.9.1 Gestalt Laws

- Common Fate
  - Things are moving together → perceptual grouped

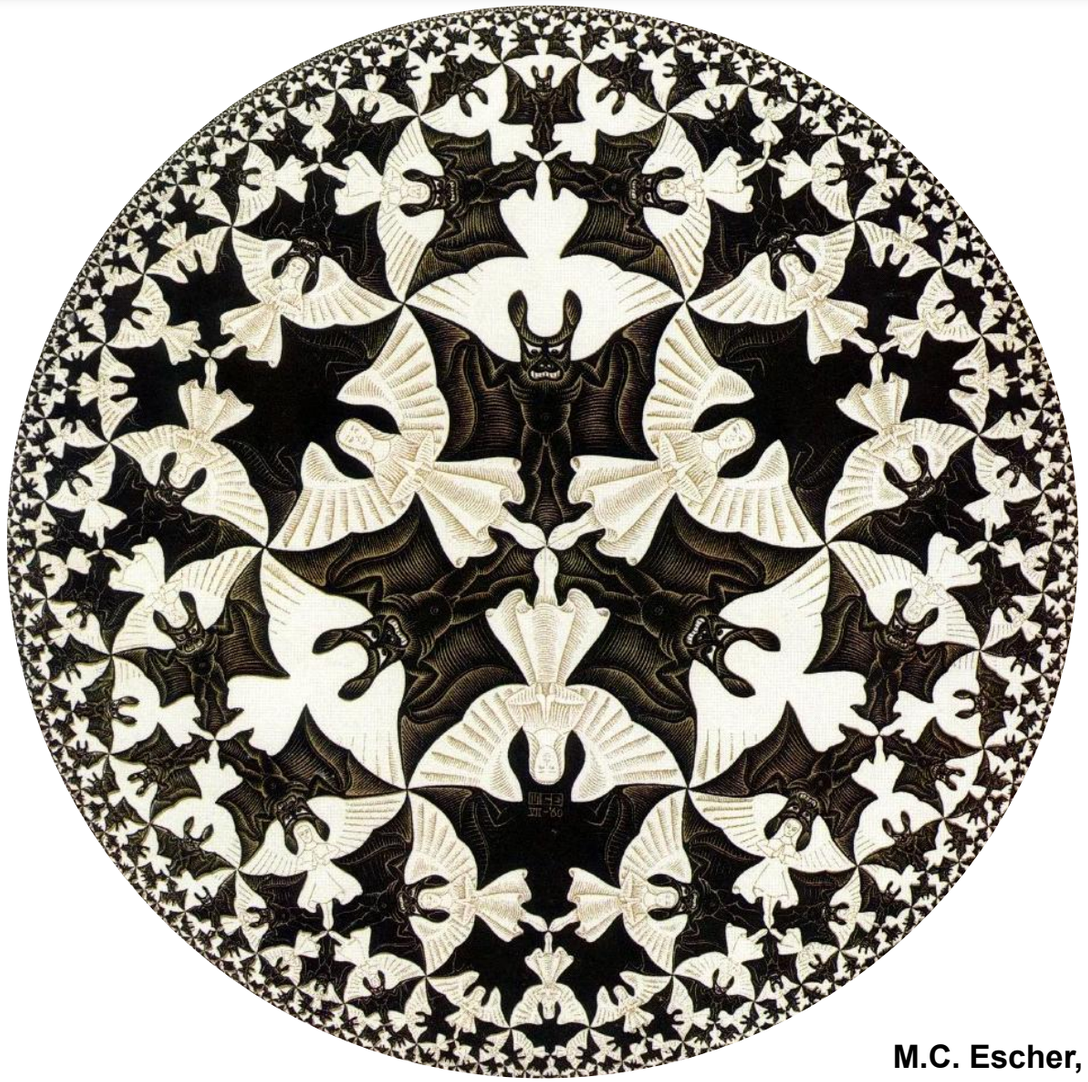


2.10. Optical illusions



What is the illusion here?

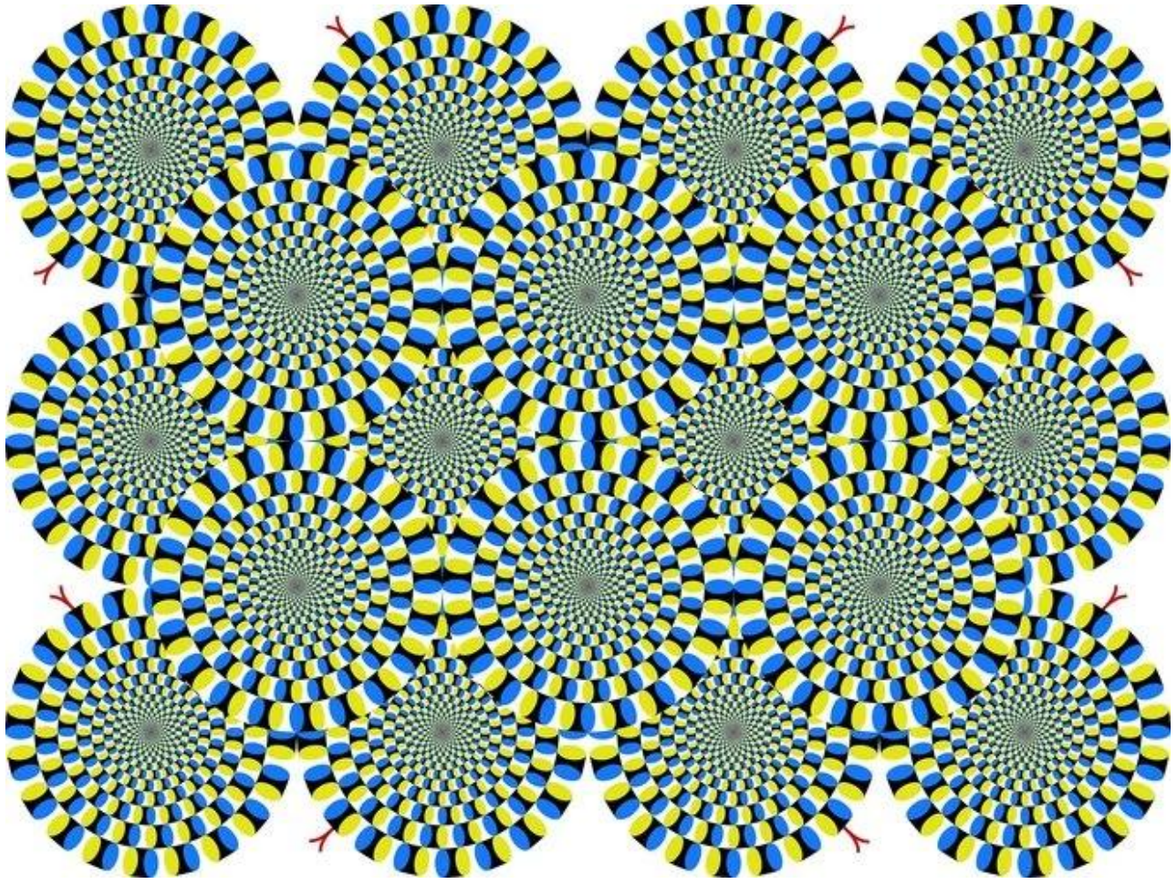
2.10. Optical illusions



M.C. Escher, Circle Limit IV (Heaven and Hell), July 1960

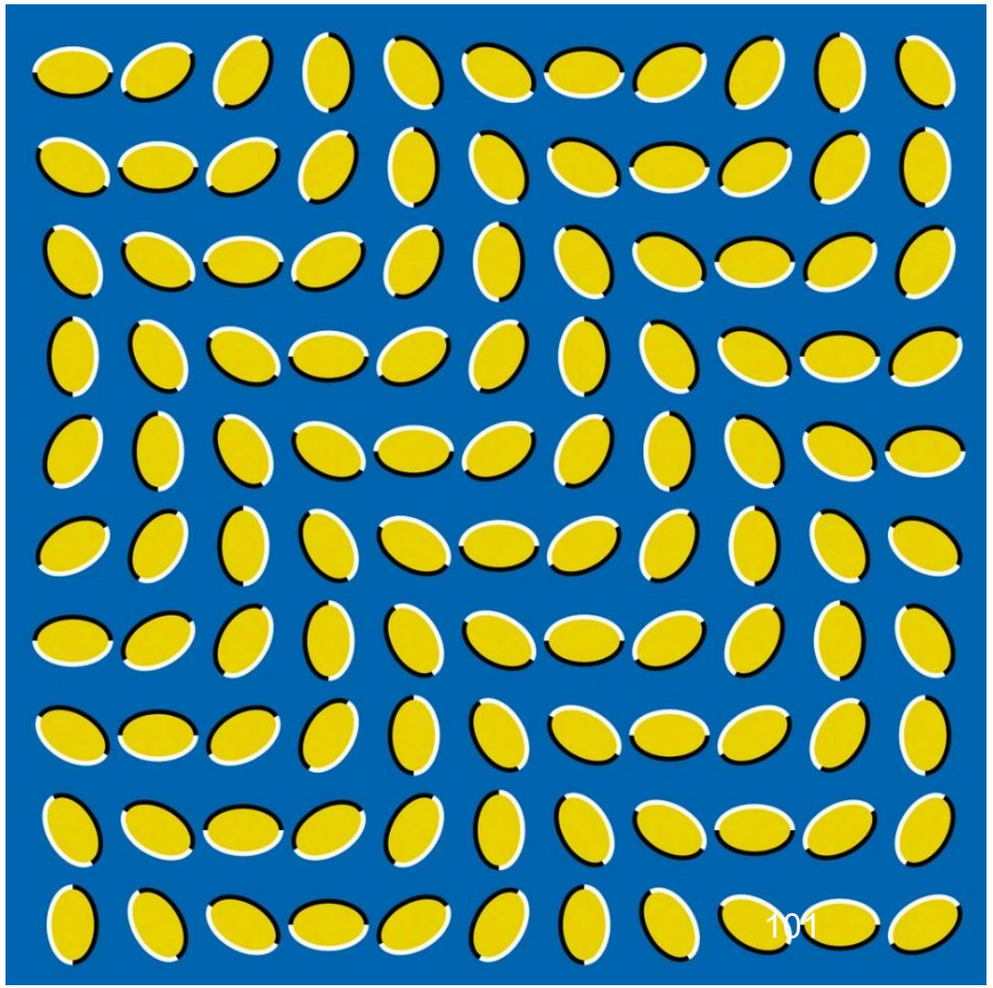
2.10. Optical illusions

Static image or animation?



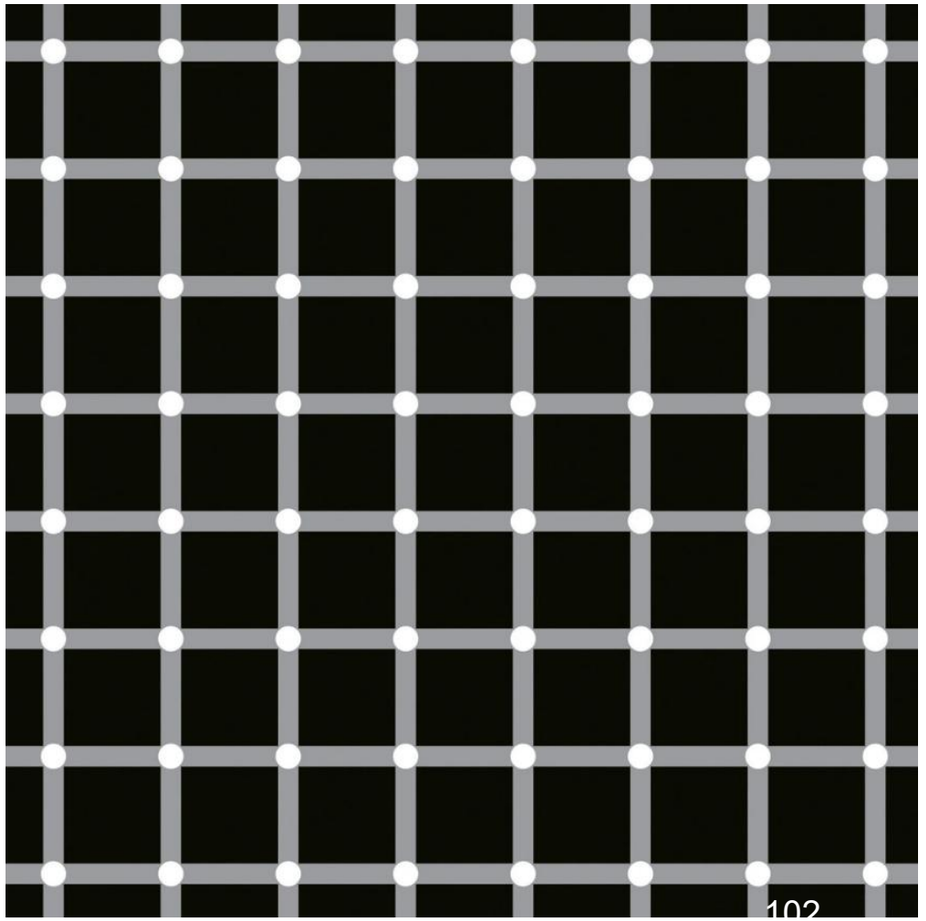
2.10. Optical illusions

Static image or animation?

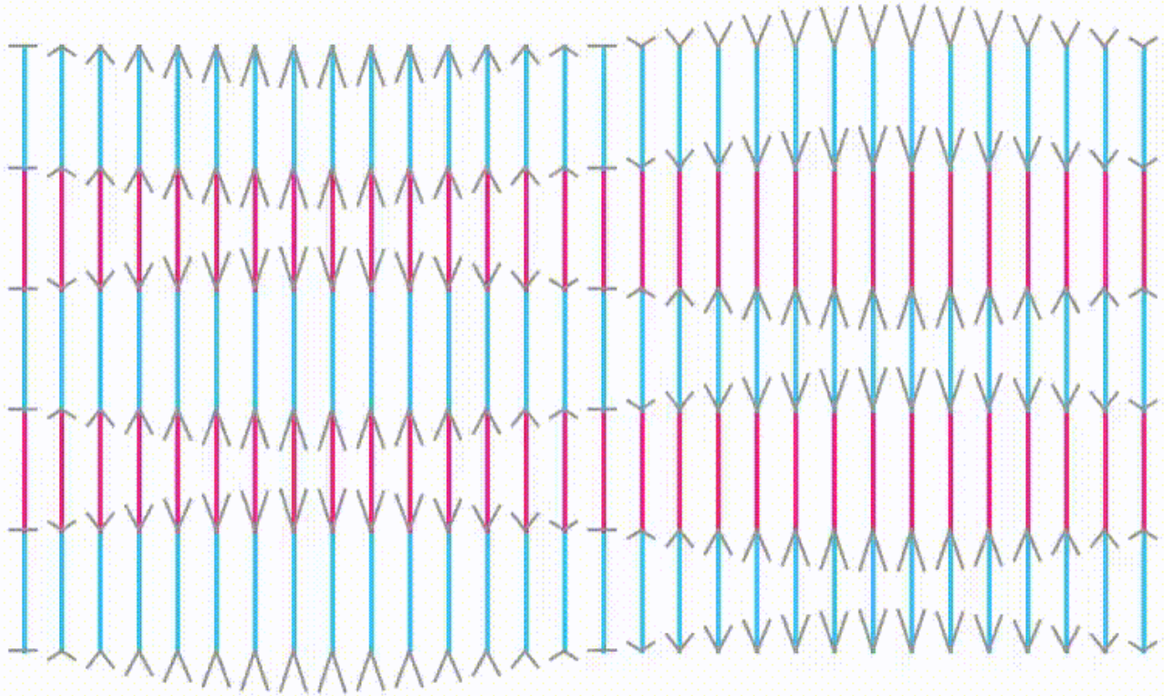


2.10. Optical illusions

Static image or animation?



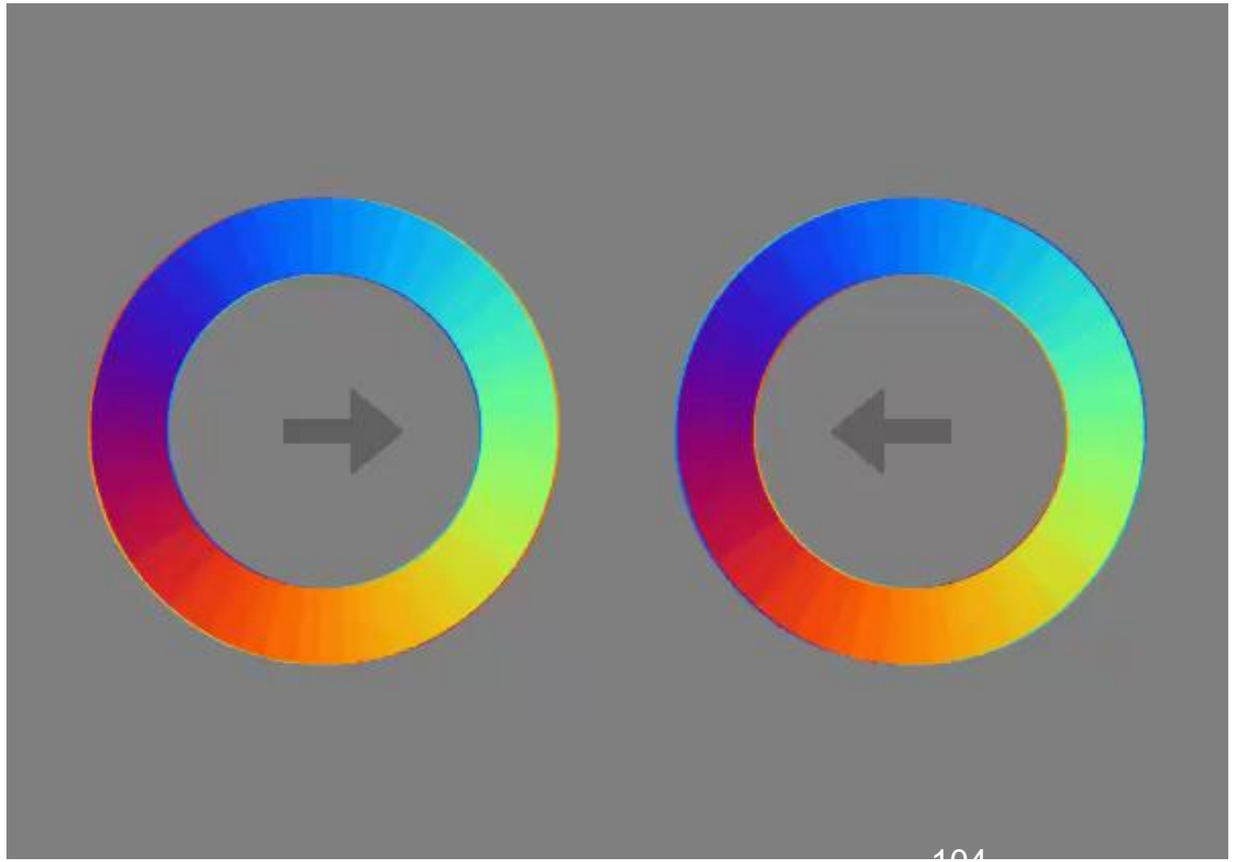
2.10. Optical illusions



None of the colored lines are moving.
Created by @Chris_Said, modified from @gsarcone

2.10. Optical illusions

- The circles don't move!



2.11. Perspective

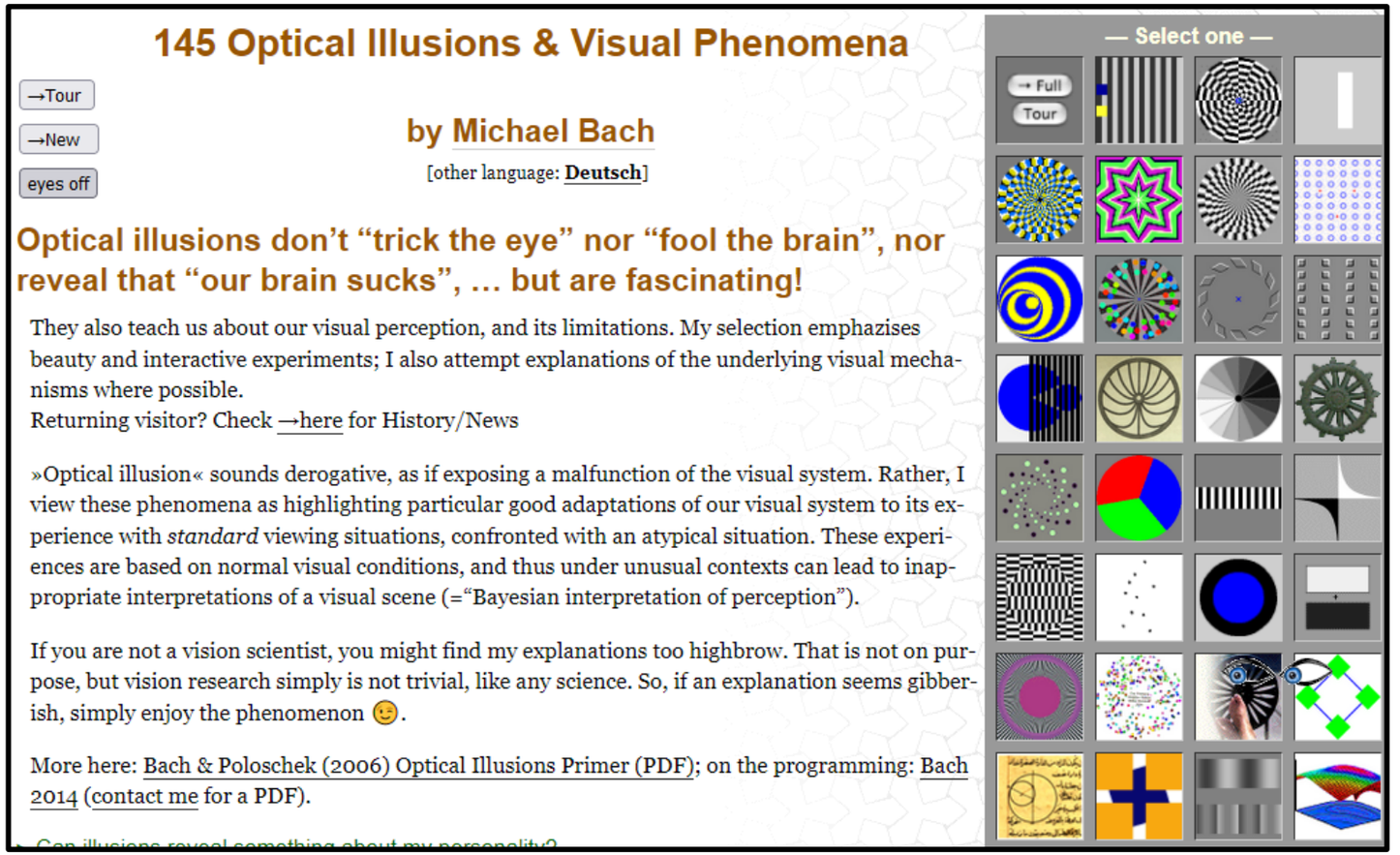
- Prof. Richard Wiseman's illusions illustrating different perspectives:



2.10. Optical illusions



Michael Bach,
Freiburg, Germany



https://michaelbach.de/ot/index.html

Further Reading

- **[Mun]** T. Munzner. Visualization Analysis and Design. 1st Edition, A K Peters/CRC Press, ISBN 978-1466508910, 2014.
  - Chapter 5: Marks and Channels

